

EDUCATION EFFECT ON PSYCHOLOGICAL TRAITS

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INTRODUCTION

There has been an increase in political polarisation across the world around socio-cultural attitudes such as immigration, gender equality, sexual minorities.

Historically these attitudes were studied by political scientists, but recent research shows that they can be predicted by some psychological traits.

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Historically these attitudes were studied by political scientists, but recent research shows that they can be predicted by some psychological traits.

We have identified 6 psychological traits of interest.

- trust
- altruism
- openness to experience
- future orientation
- universalism
- national pride

We would like to study what influences them.



Psychological
trait

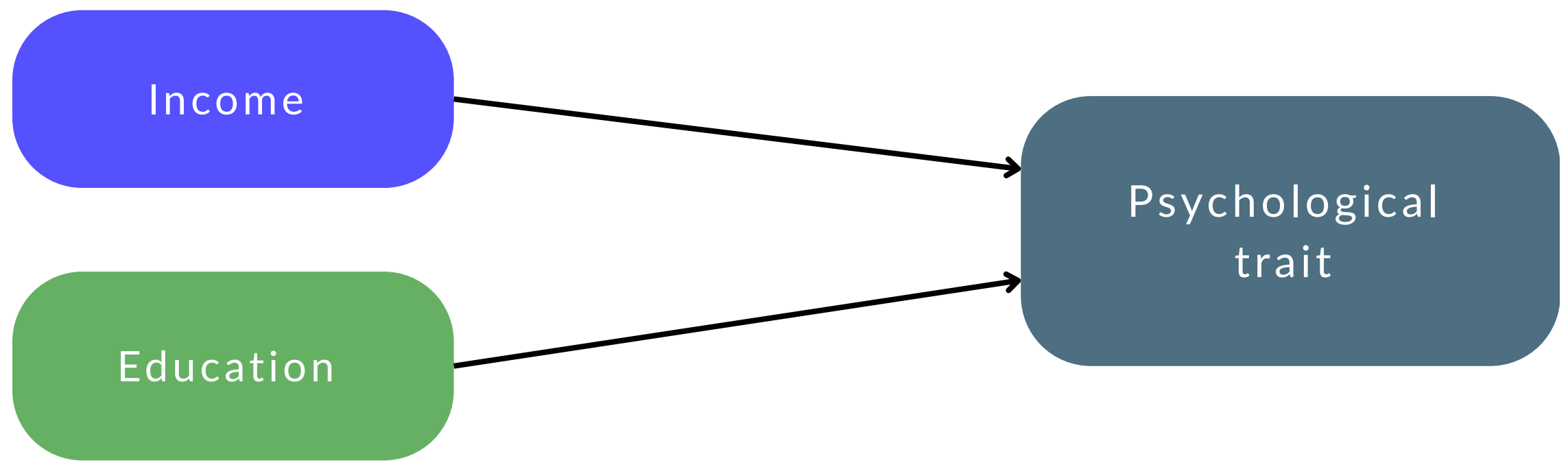


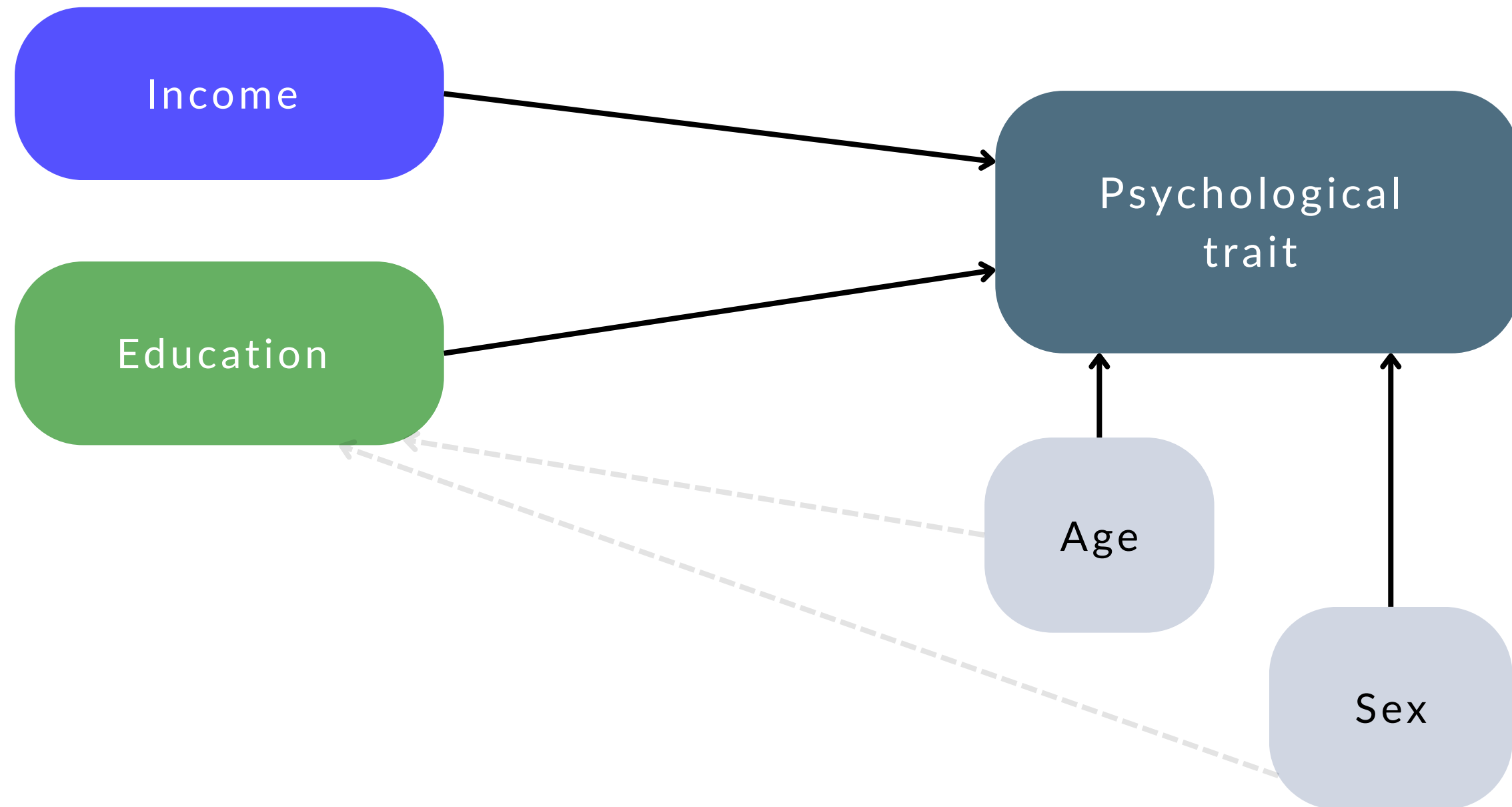
Income



Psychological
trait







H1: EDUCATION AND INCOME ARE CORRELATED WITH TRUST, ALTRUISM, OPENNESS TO EXPERIENCE, FUTURE ORIENTATION, UNIVERSALISM, NATIONAL PRIDE

DATA

EVS-WVS 2017 dataset, N = 37771 (35 countries)

DEPENDENT VARIABLES

- trust
- altruism
- openness to experience
- future orientation
- universalism
- national pride

INDEPENDENT VARIABLES

- Education
- Perceived income
- Age
- Sex

RESULTS

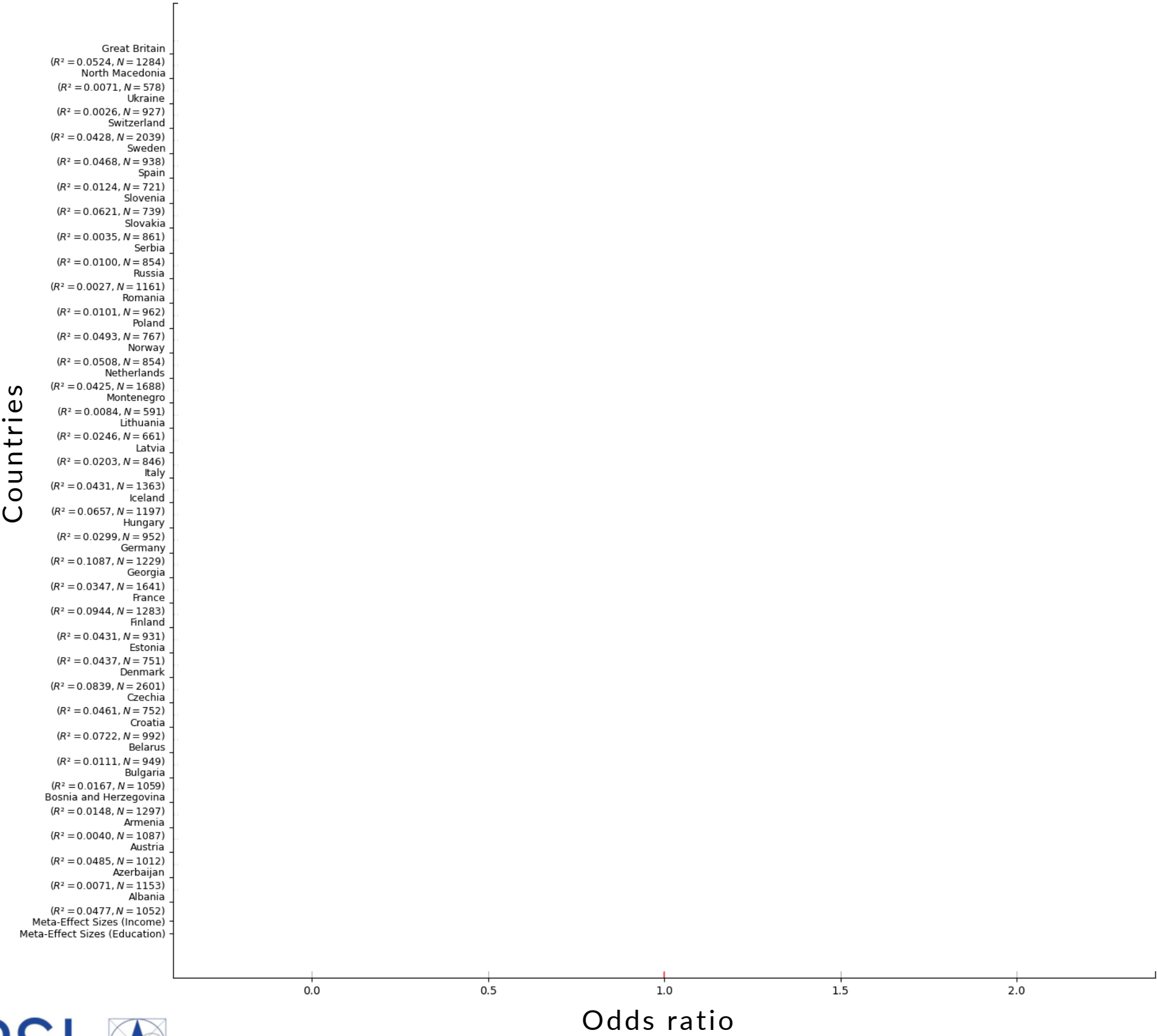
TRUST

Countries

Great Britain
($R^2 = 0.0524, N = 1284$)
North Macedonia
($R^2 = 0.0071, N = 578$)
Ukraine
($R^2 = 0.0026, N = 927$)
Switzerland
($R^2 = 0.0428, N = 2039$)
Sweden
($R^2 = 0.0468, N = 938$)
Spain
($R^2 = 0.0124, N = 721$)
Slovenia
($R^2 = 0.0621, N = 739$)
Slovakia
($R^2 = 0.0035, N = 861$)
Serbia
($R^2 = 0.0100, N = 854$)
Russia
($R^2 = 0.0027, N = 1161$)
Romania
($R^2 = 0.0101, N = 962$)
Poland
($R^2 = 0.0493, N = 767$)
Norway
($R^2 = 0.0508, N = 854$)
Netherlands
($R^2 = 0.0425, N = 1688$)
Montenegro
($R^2 = 0.0084, N = 591$)
Lithuania
($R^2 = 0.0246, N = 661$)
Latvia
($R^2 = 0.0203, N = 846$)
Italy
($R^2 = 0.0431, N = 1363$)
Iceland
($R^2 = 0.0657, N = 1197$)
Hungary
($R^2 = 0.0299, N = 952$)
Germany
($R^2 = 0.1087, N = 1229$)
Georgia
($R^2 = 0.0347, N = 1641$)
France
($R^2 = 0.0944, N = 1283$)
Finland
($R^2 = 0.0431, N = 931$)
Estonia
($R^2 = 0.0437, N = 751$)
Denmark
($R^2 = 0.0839, N = 2601$)
Czechia
($R^2 = 0.0461, N = 752$)
Croatia
($R^2 = 0.0722, N = 992$)
Belarus
($R^2 = 0.0111, N = 949$)
Bulgaria
($R^2 = 0.0167, N = 1059$)
Bosnia and Herzegovina
($R^2 = 0.0148, N = 1297$)
Armenia
($R^2 = 0.0040, N = 1087$)
Austria
($R^2 = 0.0485, N = 1012$)
Azerbaijan
($R^2 = 0.0071, N = 1153$)
Albania
($R^2 = 0.0477, N = 1052$)
Meta-Effect Sizes (Income)
Meta-Effect Sizes (Education)

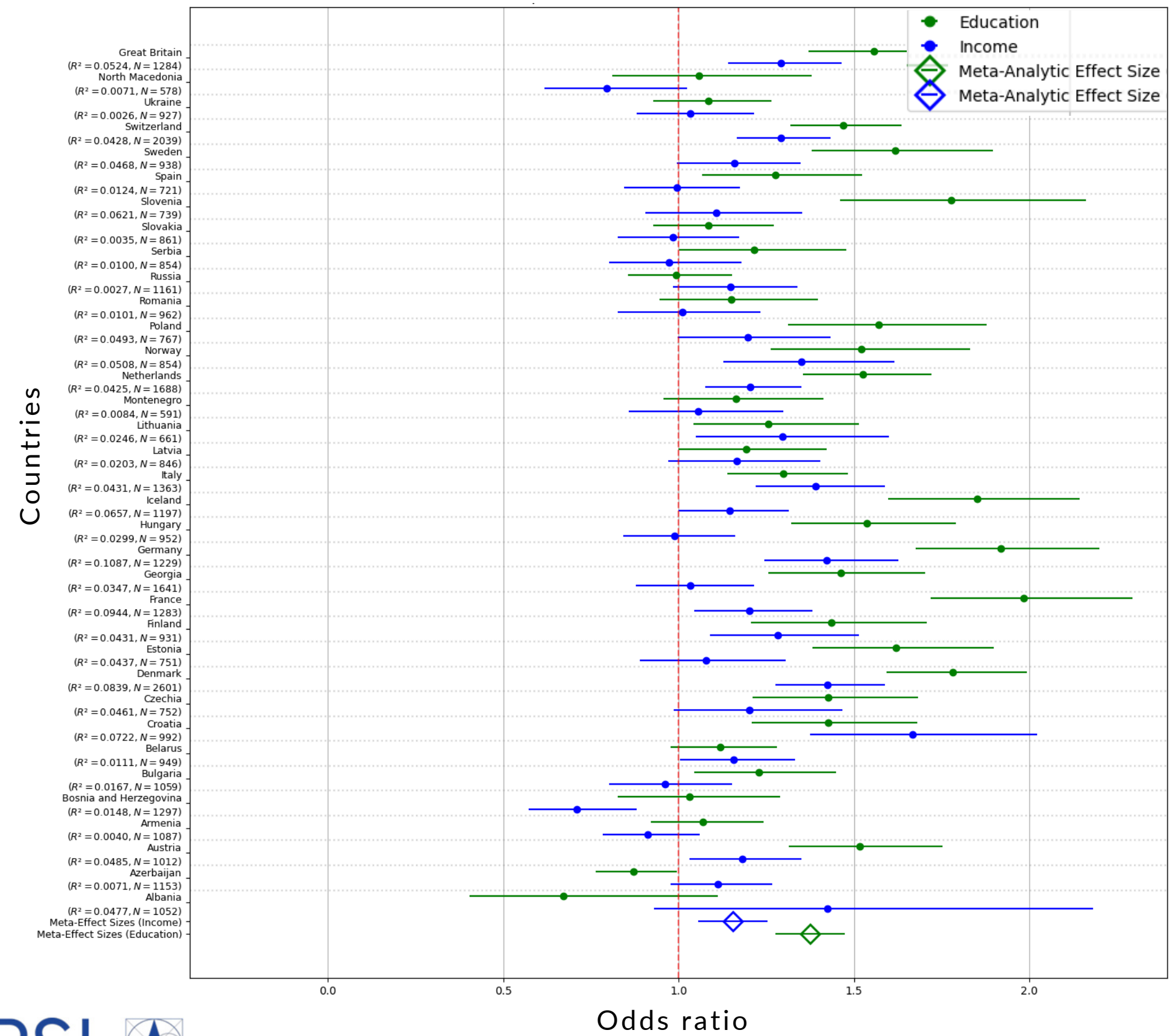
RESULTS

TRUST



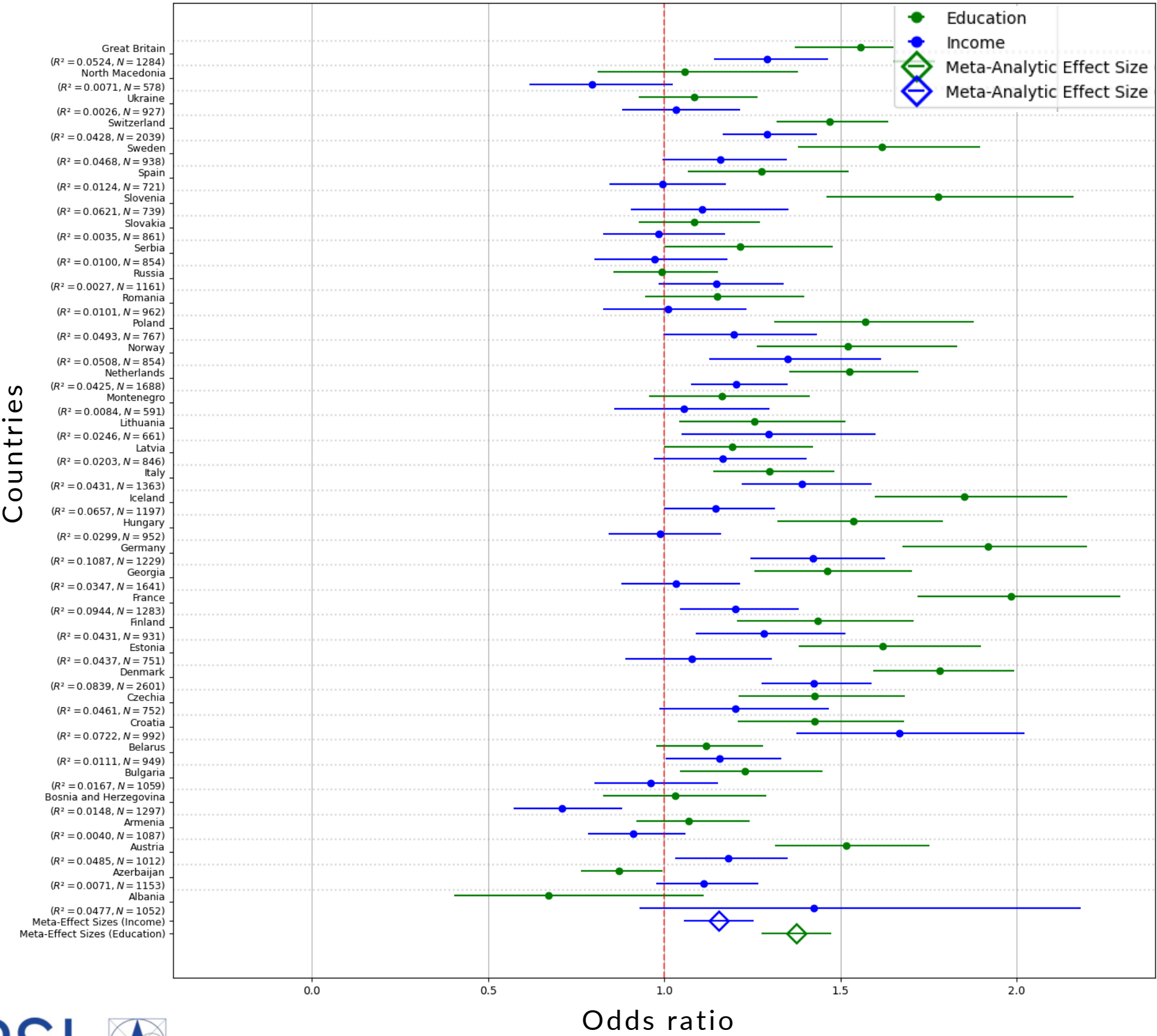
RESULTS

TRUST

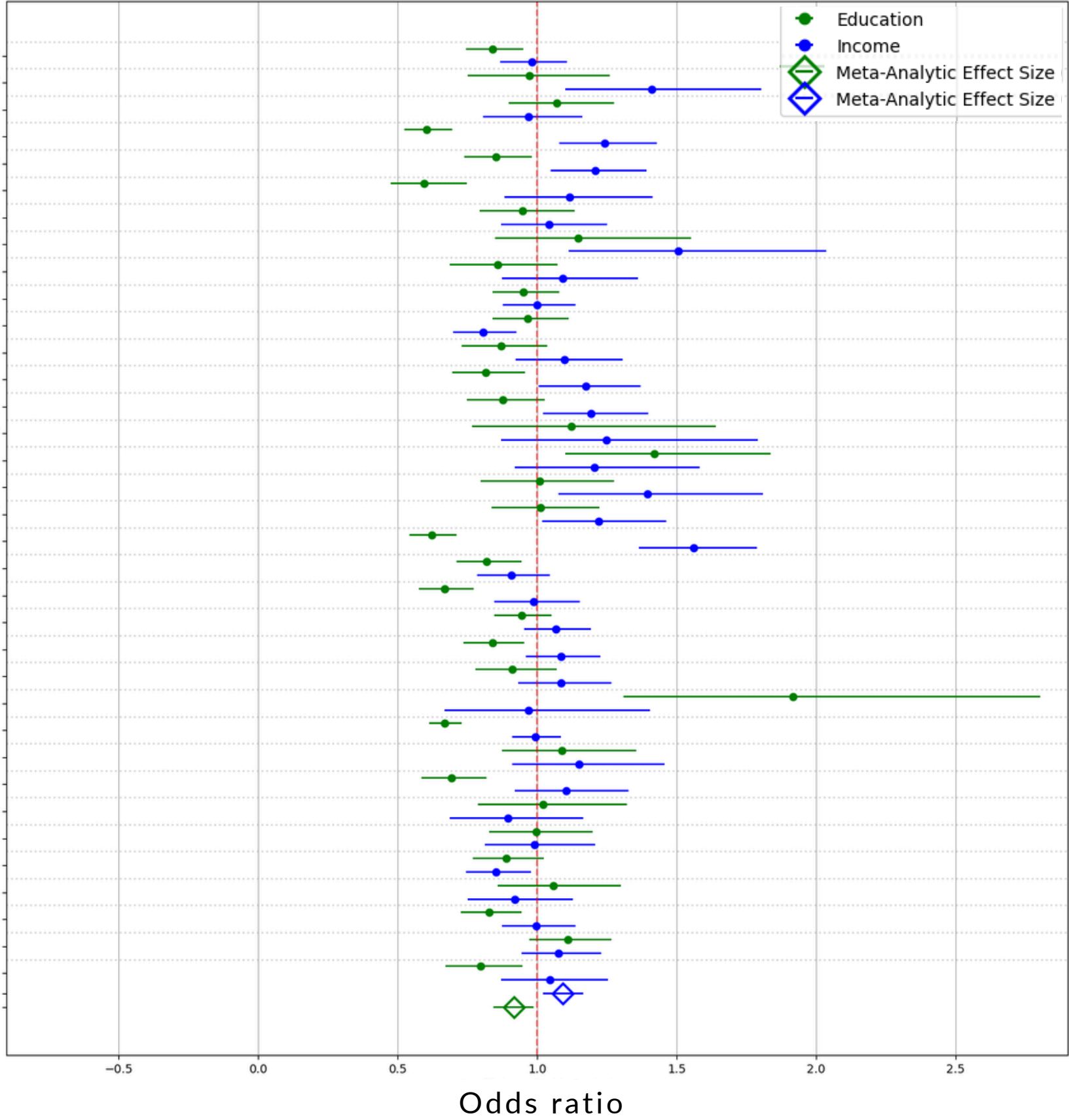


RESULTS

TRUST



NATIONAL PRIDE



HETEROGENEITY

H2 : THERE IS A DIFFERENCE IN EDUCATION EFFECT BETWEEN COUNTRIES

HETEROGENEITY

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Cochran's Q and I^2 test

Variable	I^2 education
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Trust	
-------	--

Altruism	
----------	--

Openness to exp	
-----------------	--

Future orientation	
--------------------	--

Universalism	
--------------	--

National pride	
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HETEROGENEITY

H2 : THERE IS A DIFFERENCE IN EDUCATION EFFECT BETWEEN COUNTRIES

Cochran's Q and I^2 test

Variable	I^2 education
----------	-----------------

Trust	85.97%
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Altruism	
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Openness to exp	
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Future orientation	
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Universalism	
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National pride	
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Cochran's Q and I^2 test

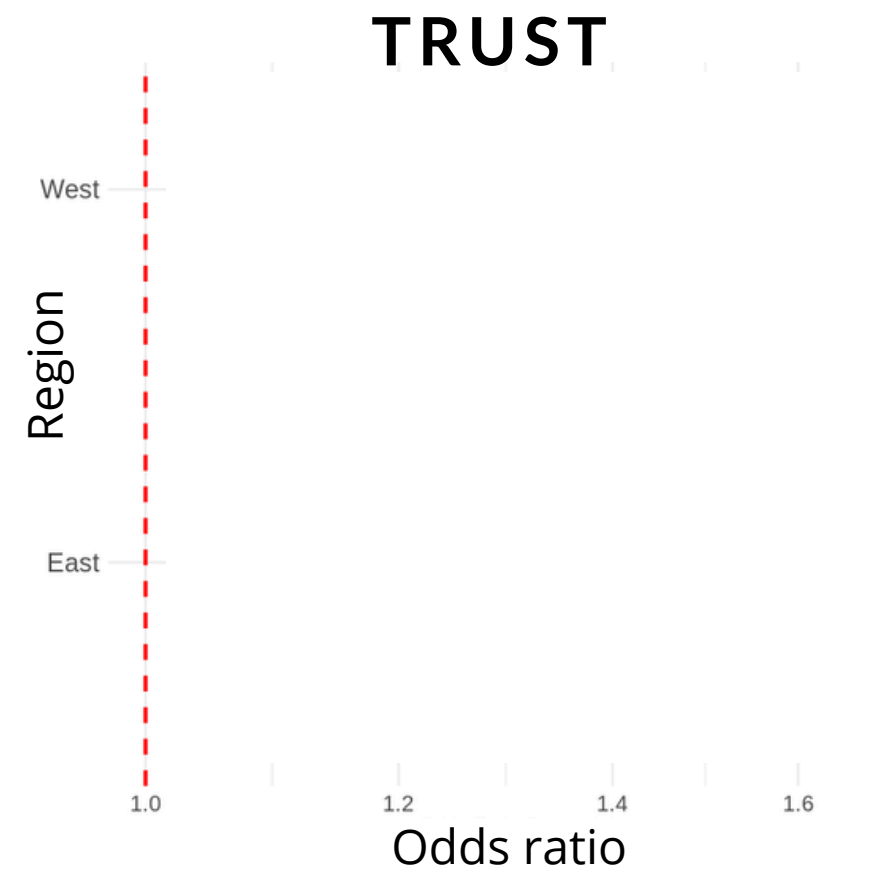
Variable	I^2 education
Trust	85.97%
Altruism	50.09%
Openness to exp	76.47%
Future orientation	70.36%
Universalism	67.60%
National pride	83.74%

EAST VS WEST? EDUCATION

H3: THERE IS A DIFFERENCE IN EDUCATION EFFECT BETWEEN EAST AND WEST

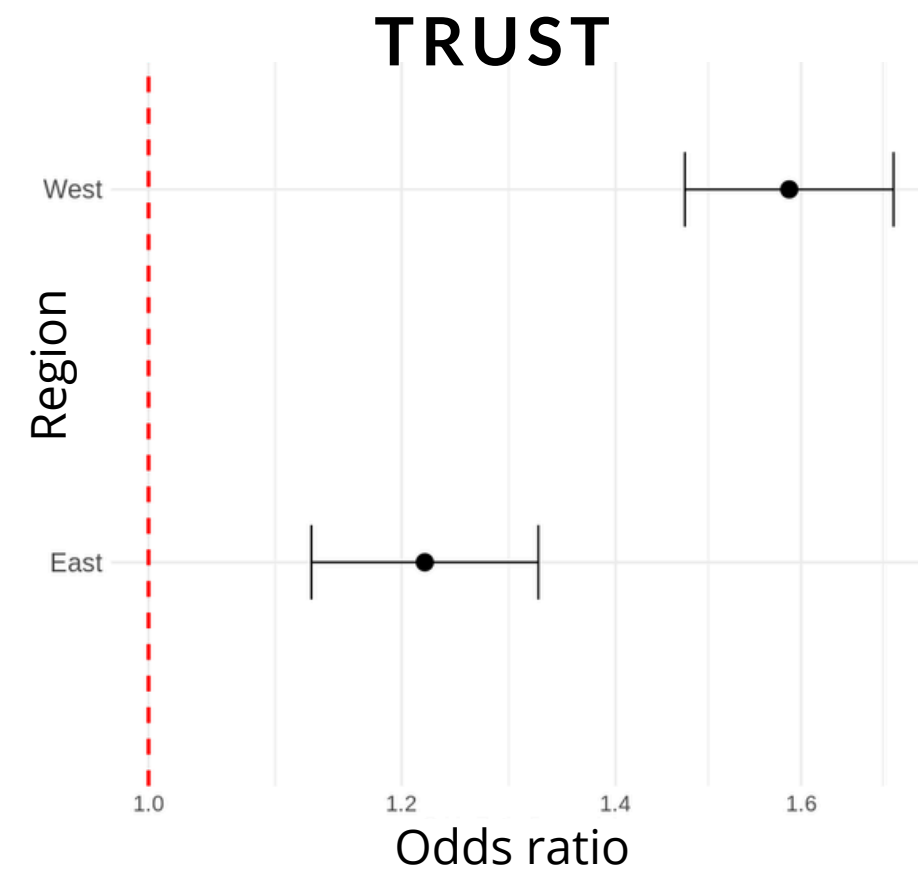
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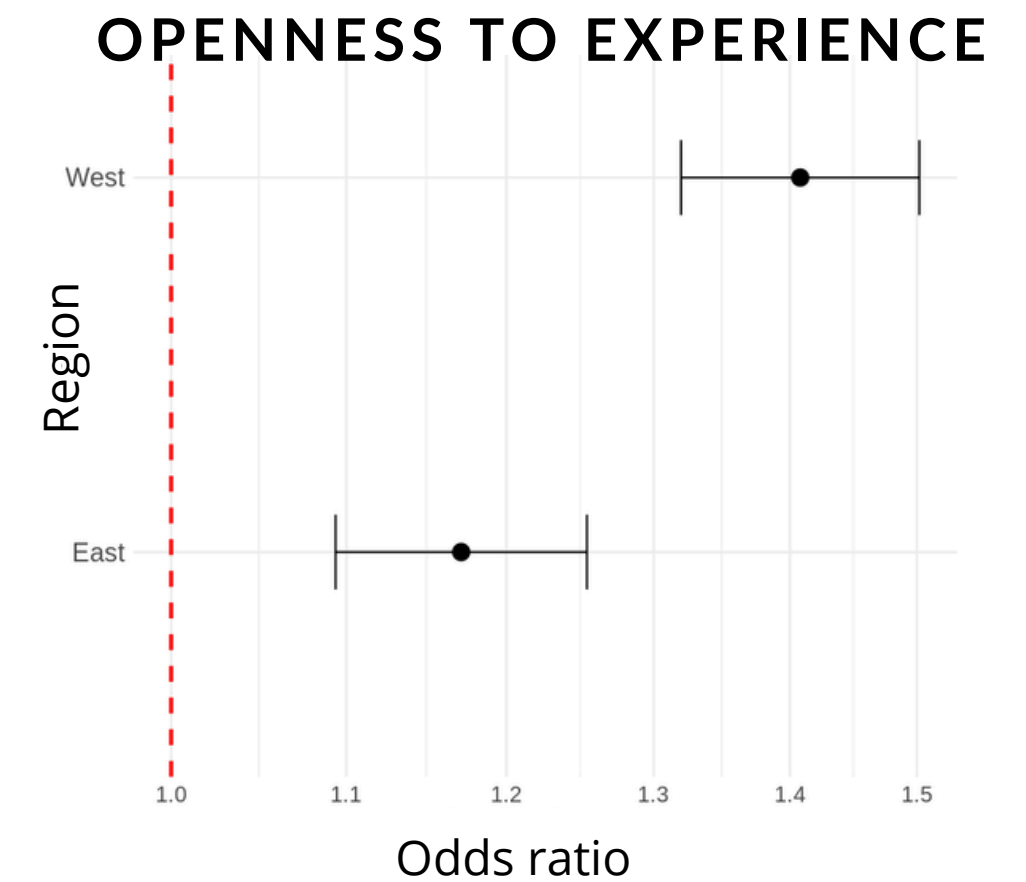
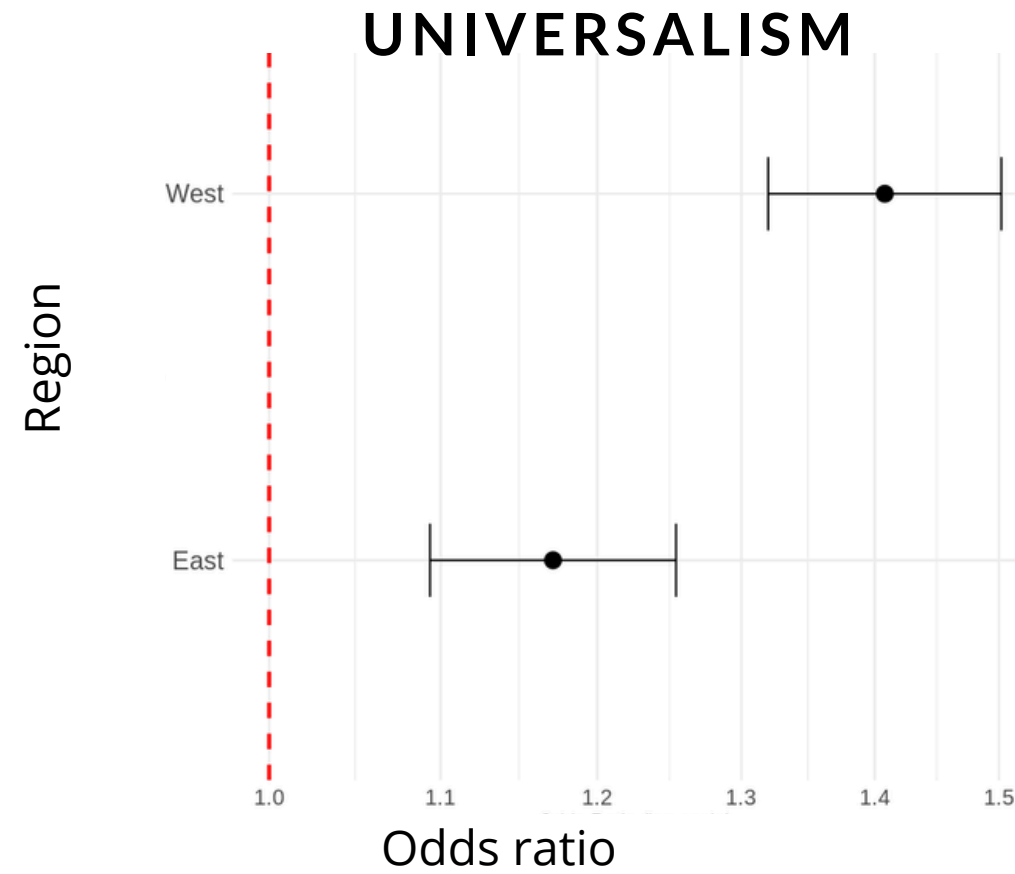
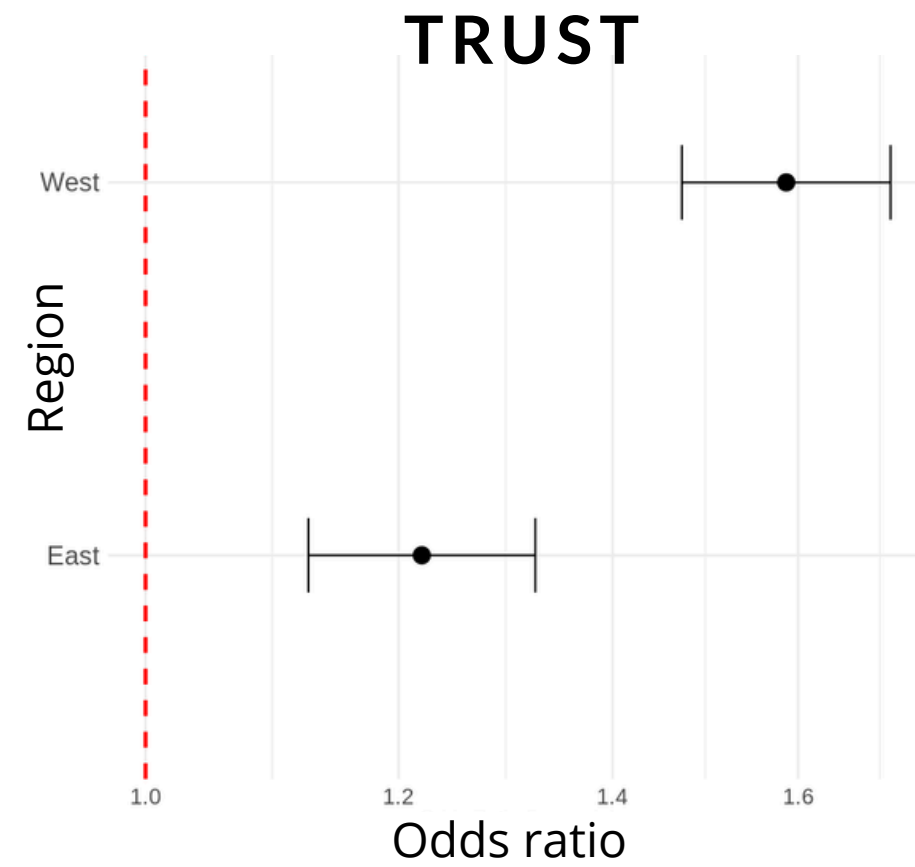
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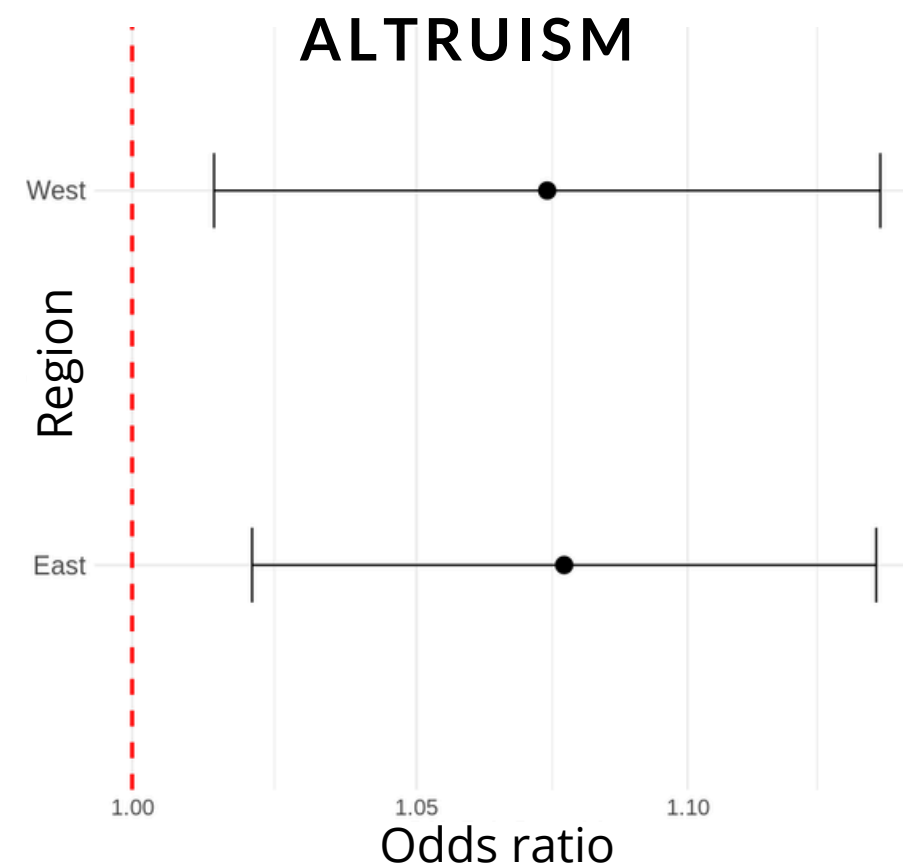
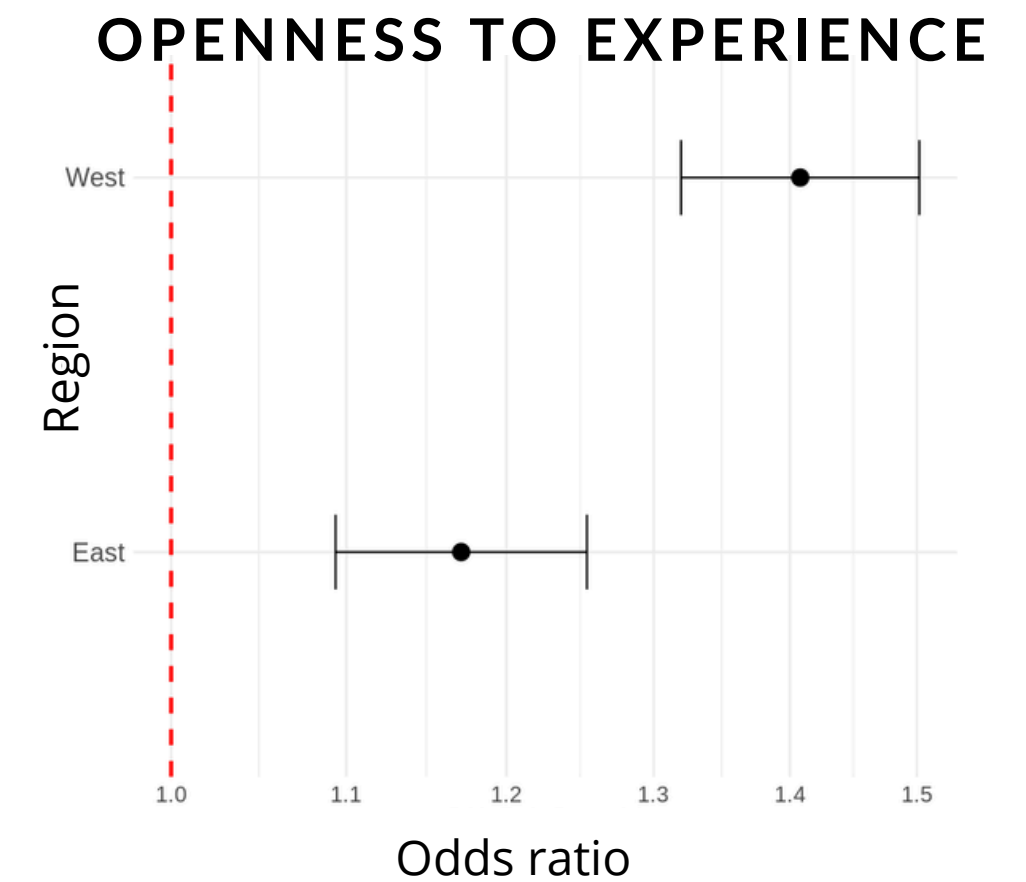
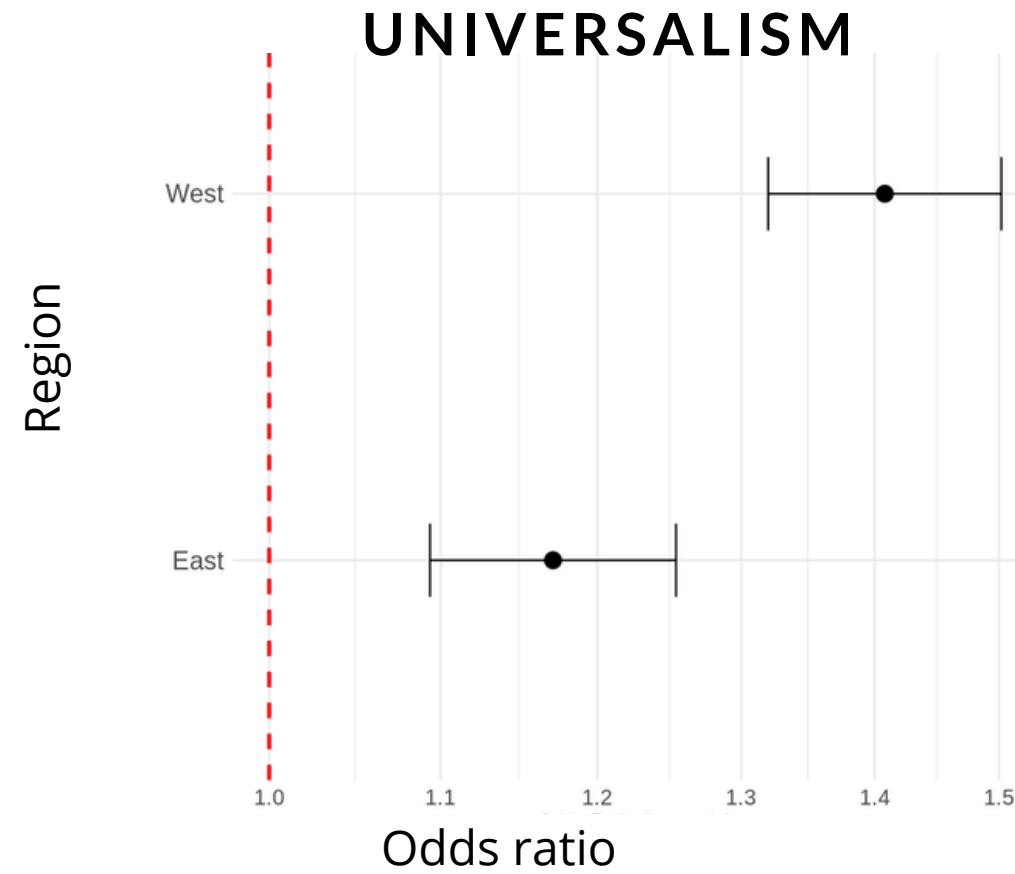
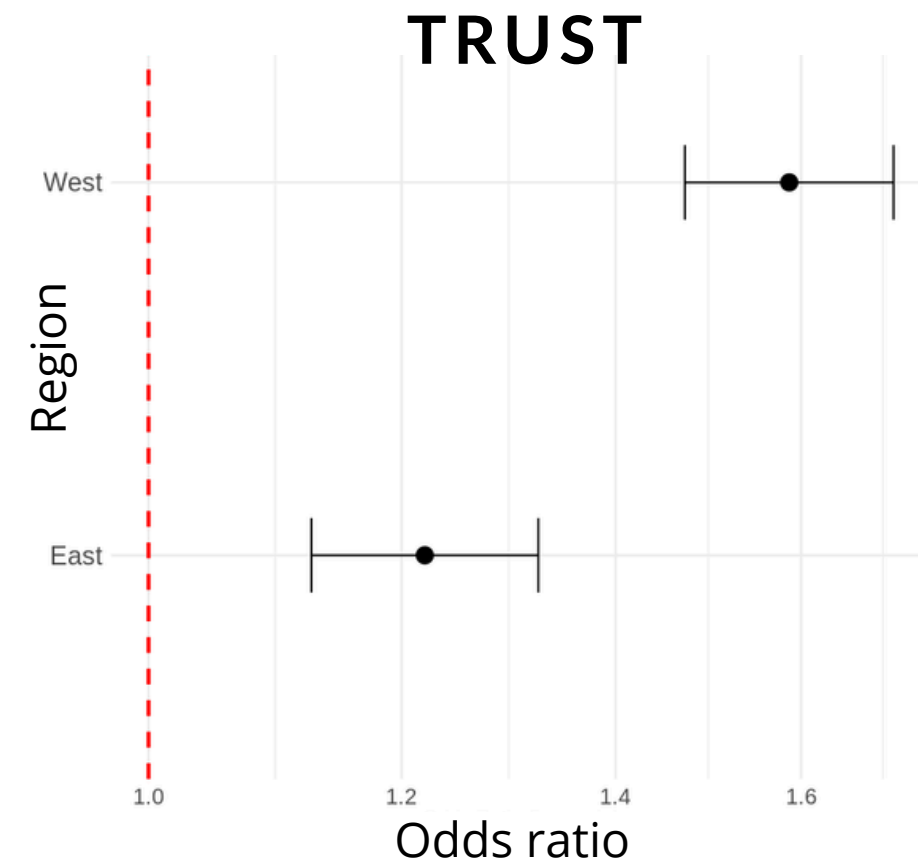
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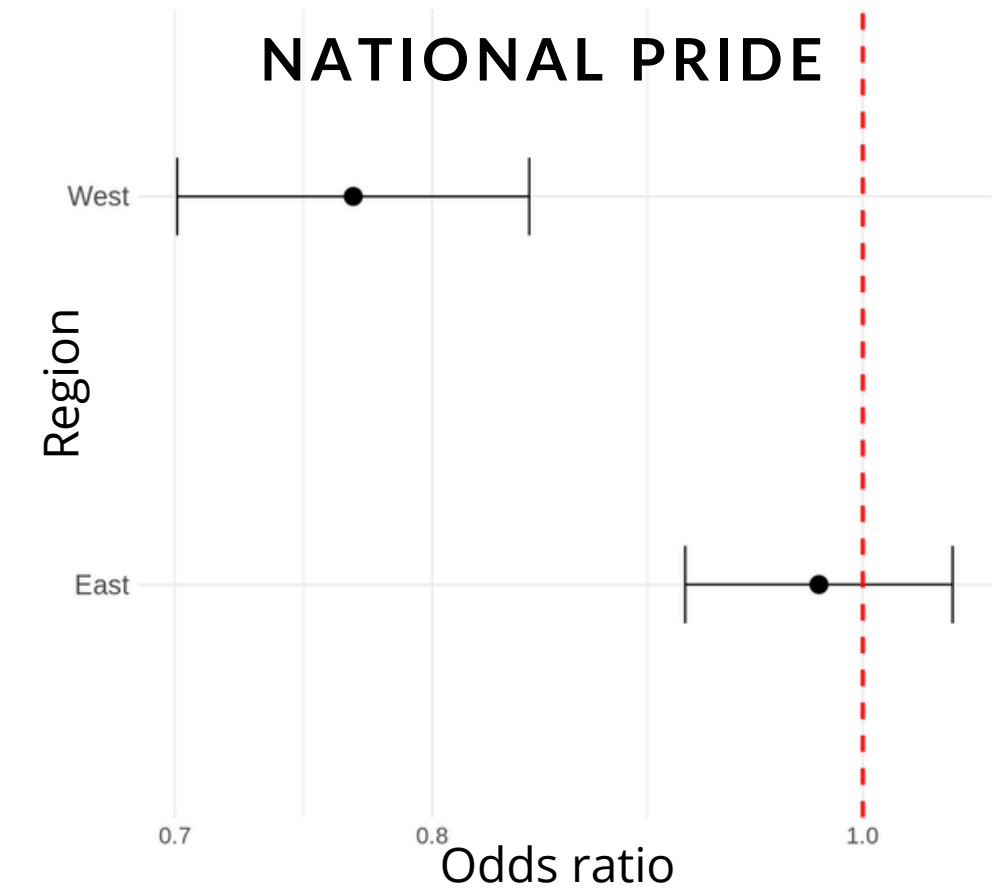
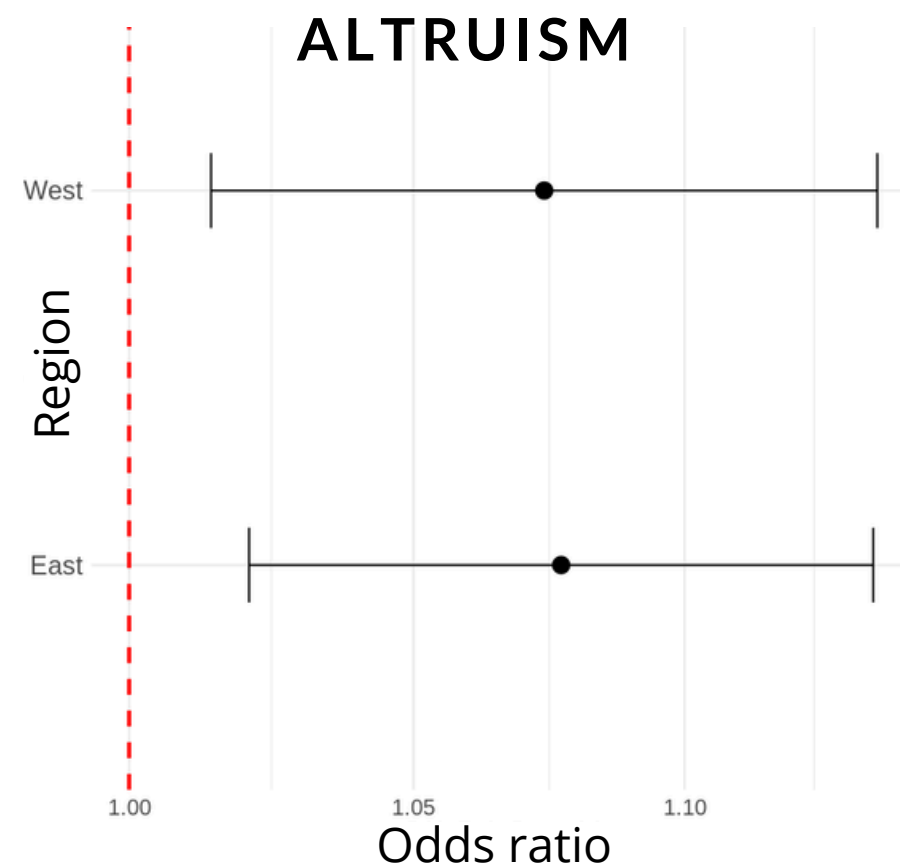
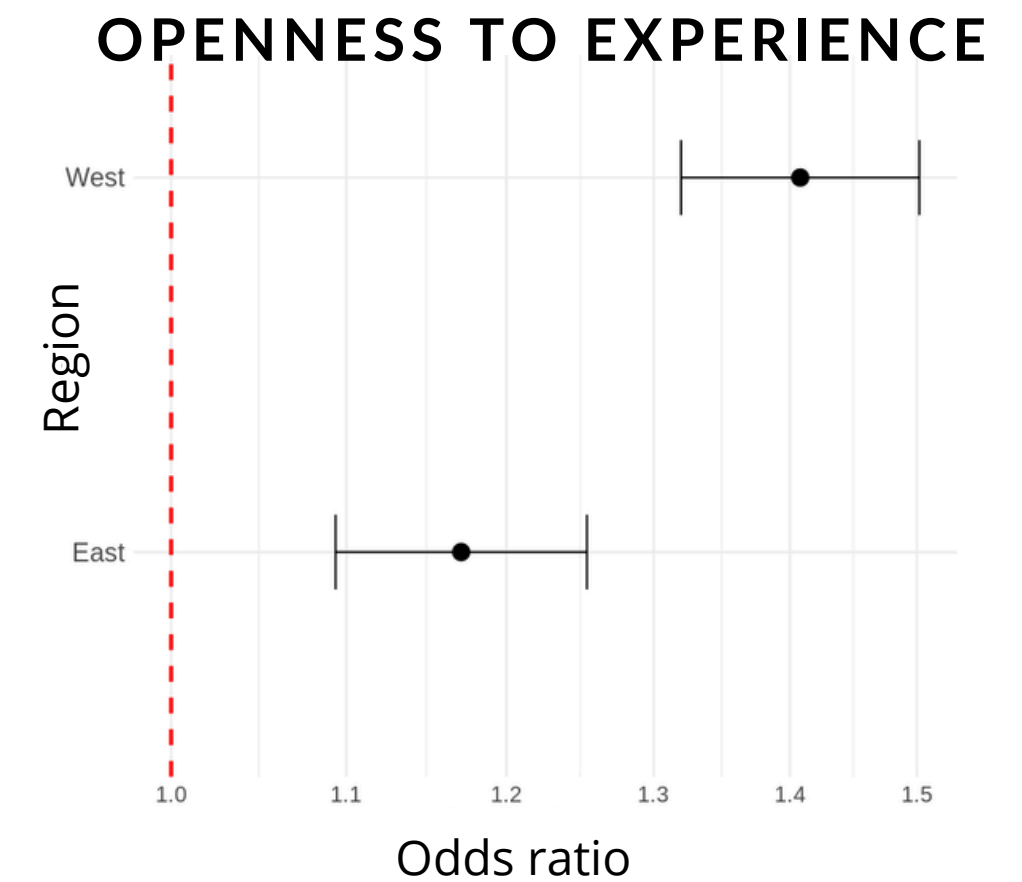
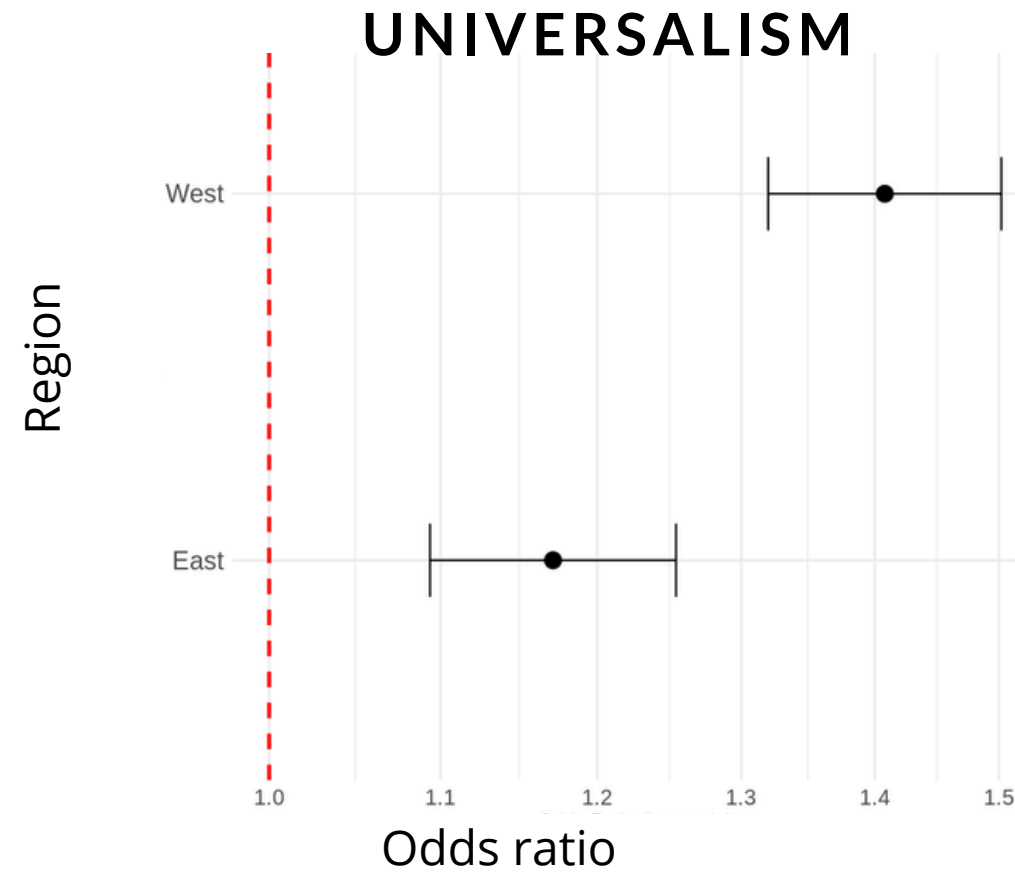
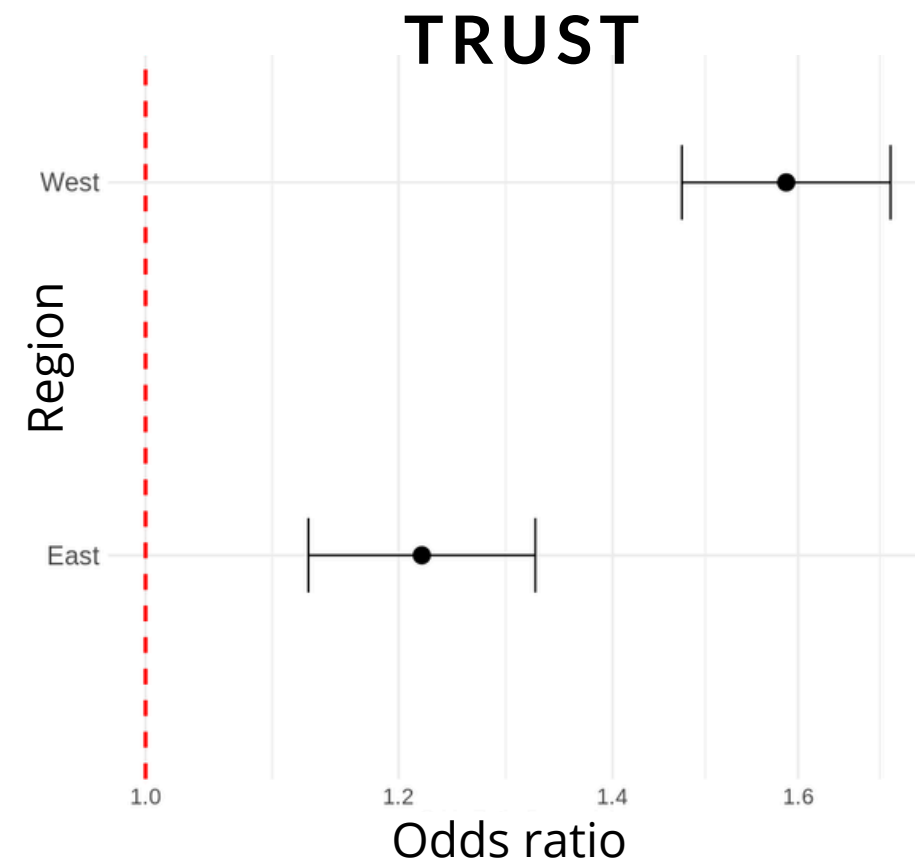
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■ GENERATIONS

H4: THERE IS A DIFFERENCE IN EDUCATION EFFECT BETWEEN GENERATIONS

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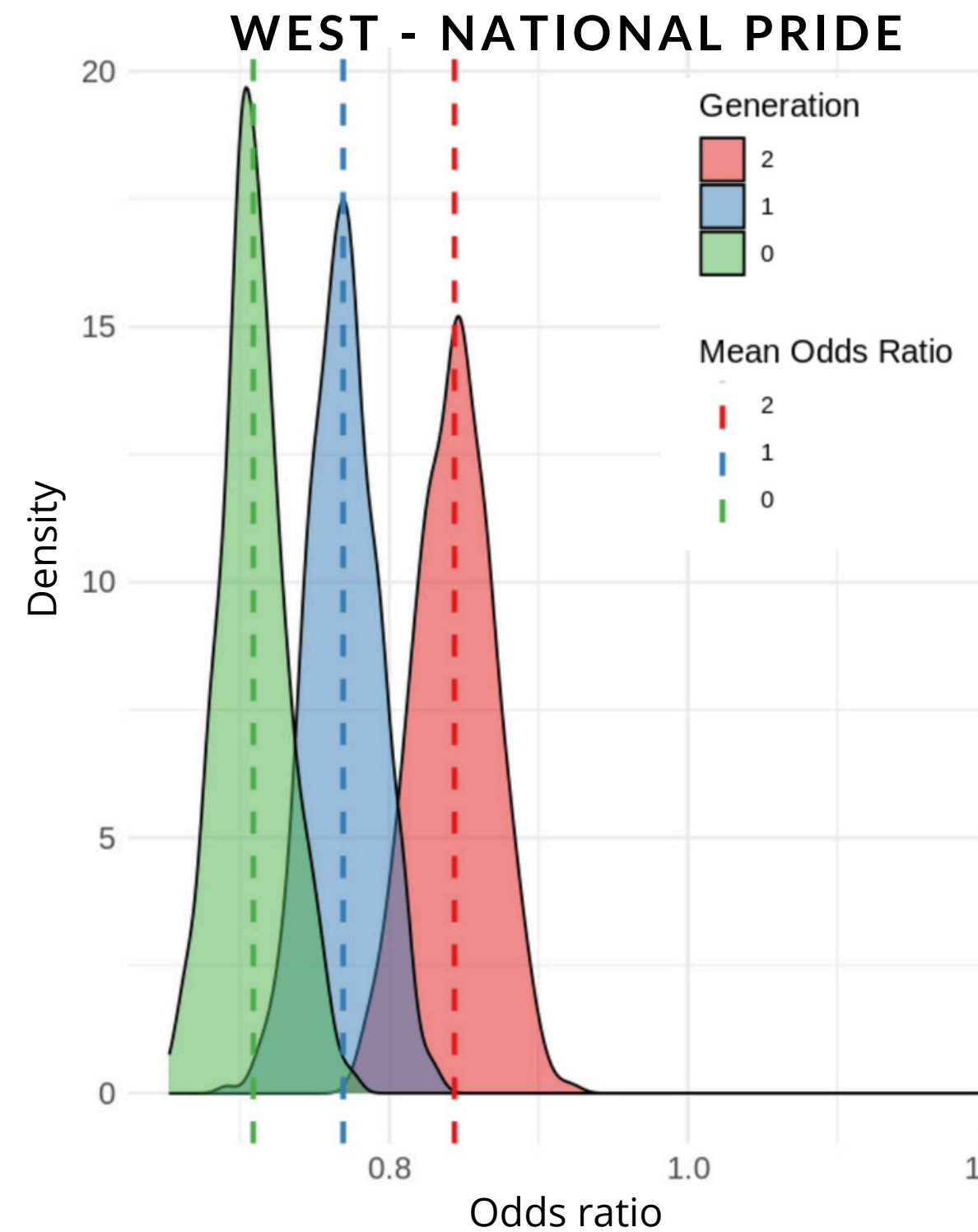
WEST - NATIONAL PRIDE

Density

Odds ratio

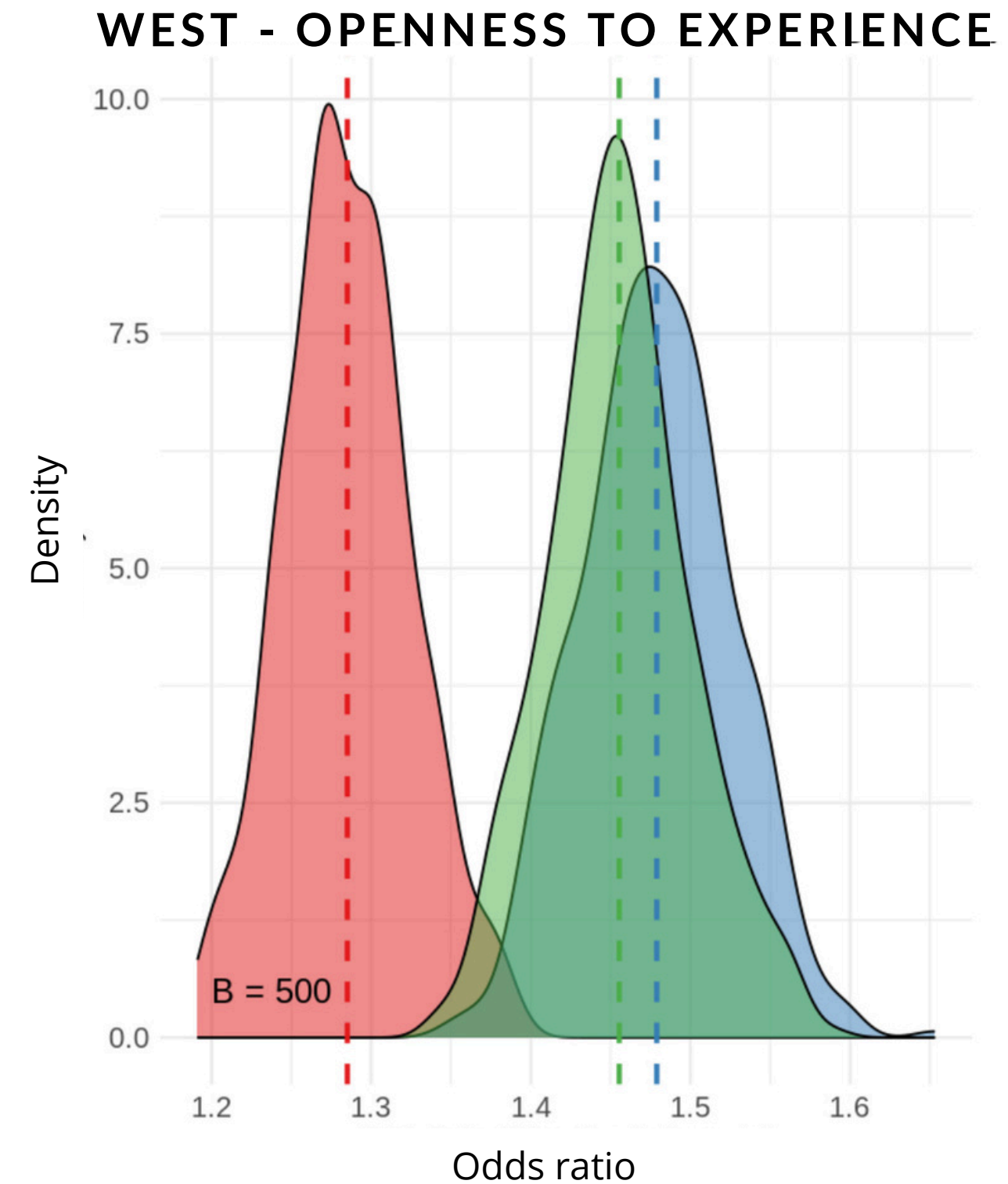
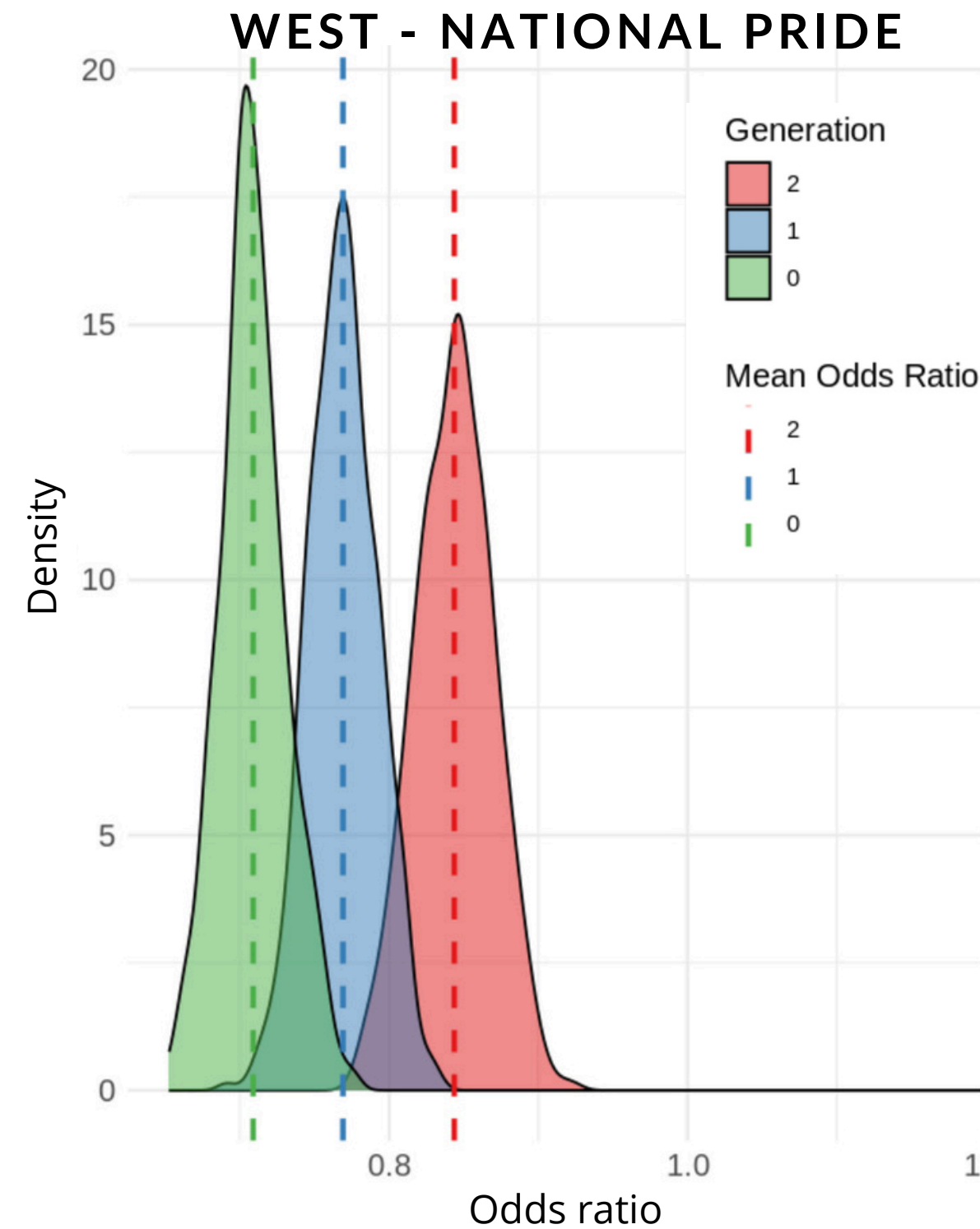
■ GENERATIONS

H4: THERE IS A DIFFERENCE IN EDUCATION EFFECT BETWEEN GENERATIONS



■ GENERATIONS

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■ GENERAL EDUCATION LEVEL

H5: GENERAL EDUCATION HAS AN EFFECT ON EDUCATION EFFECT

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H5: GENERAL EDUCATION HAS AN EFFECT ON EDUCATION EFFECT

Education

Income

Age

Male

GDP

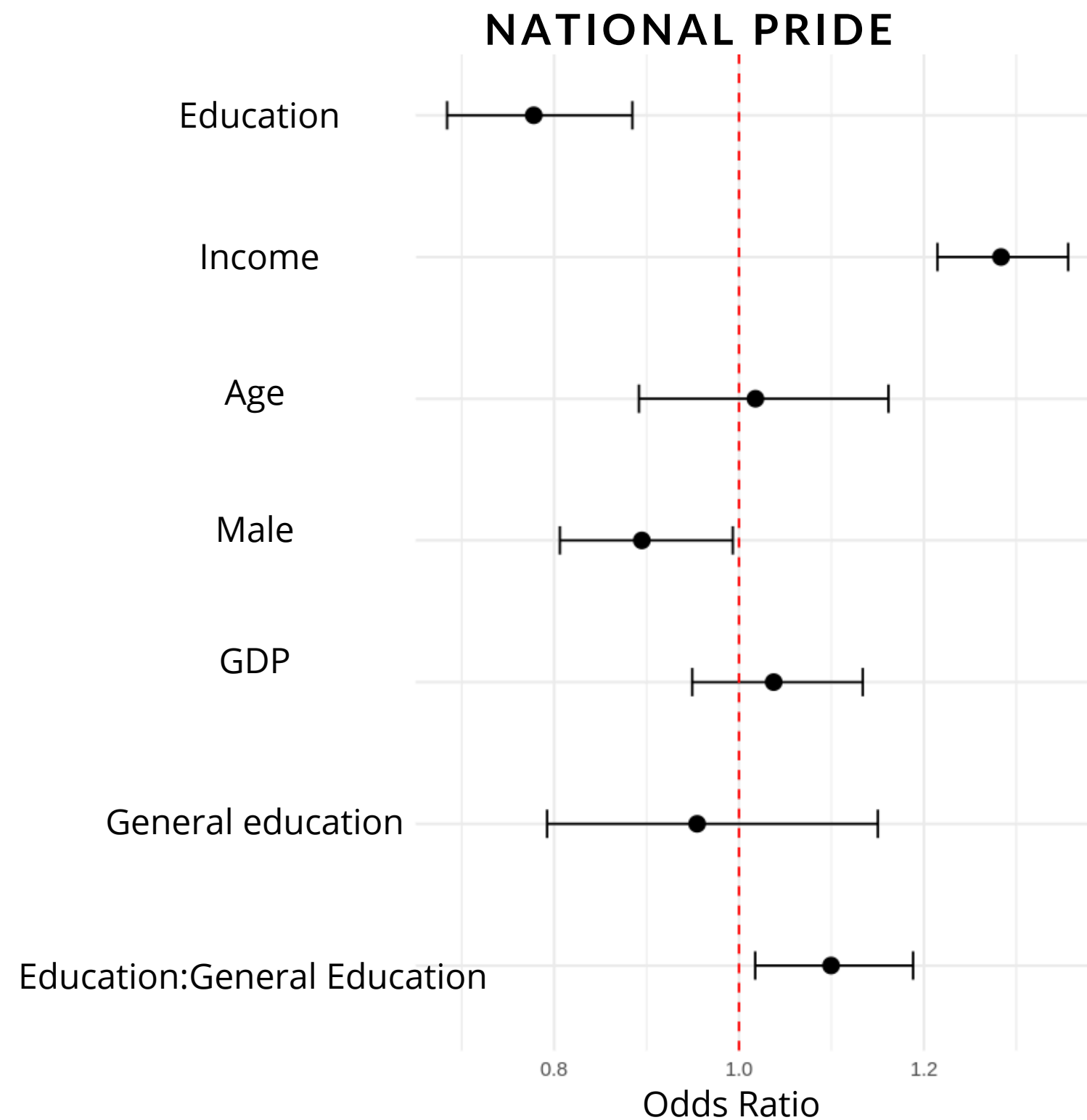
General education

Education:General Education

Odds Ratio

■ GENERAL EDUCATION LEVEL

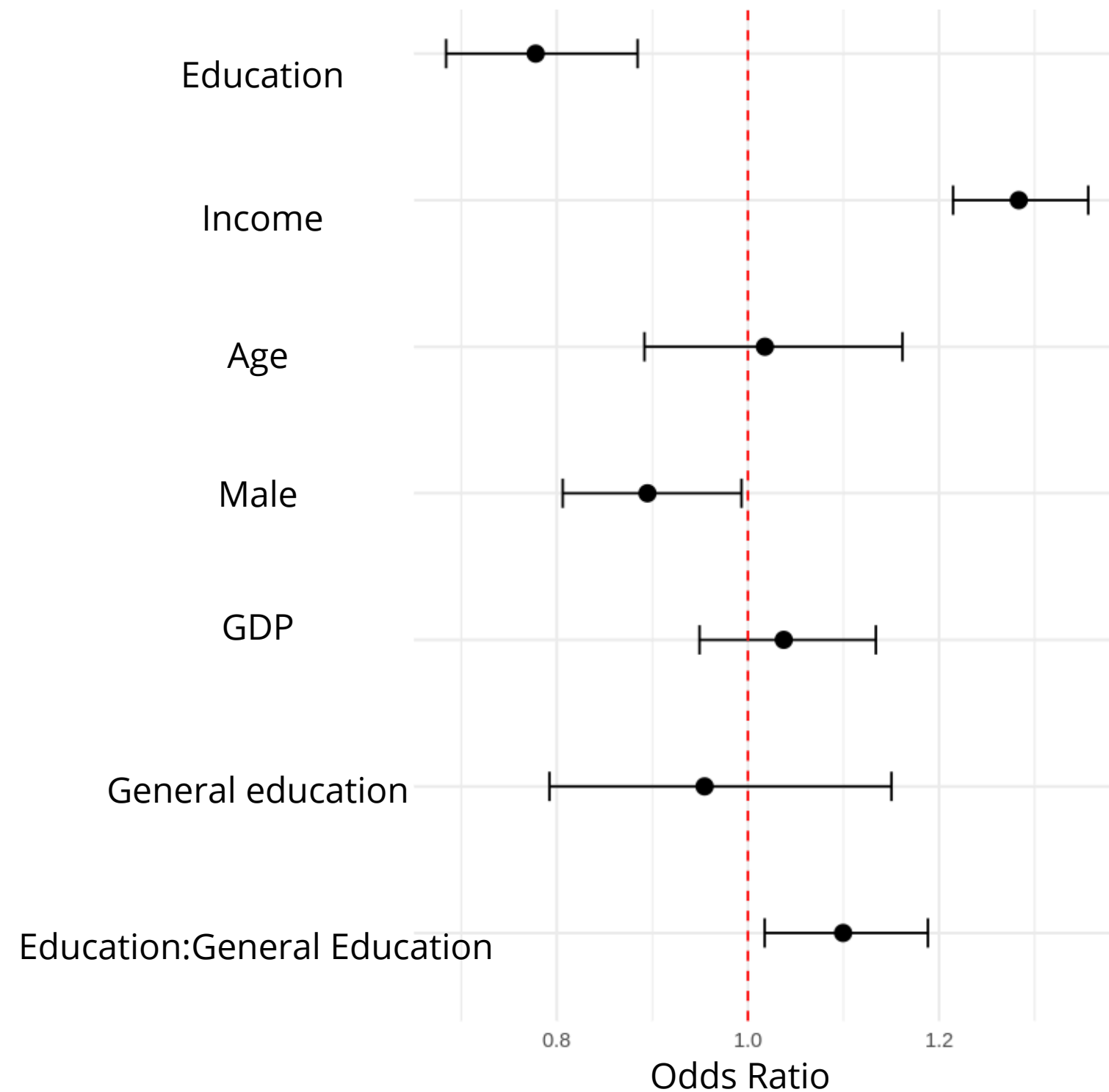
H5: GENERAL EDUCATION HAS AN EFFECT ON EDUCATION EFFECT



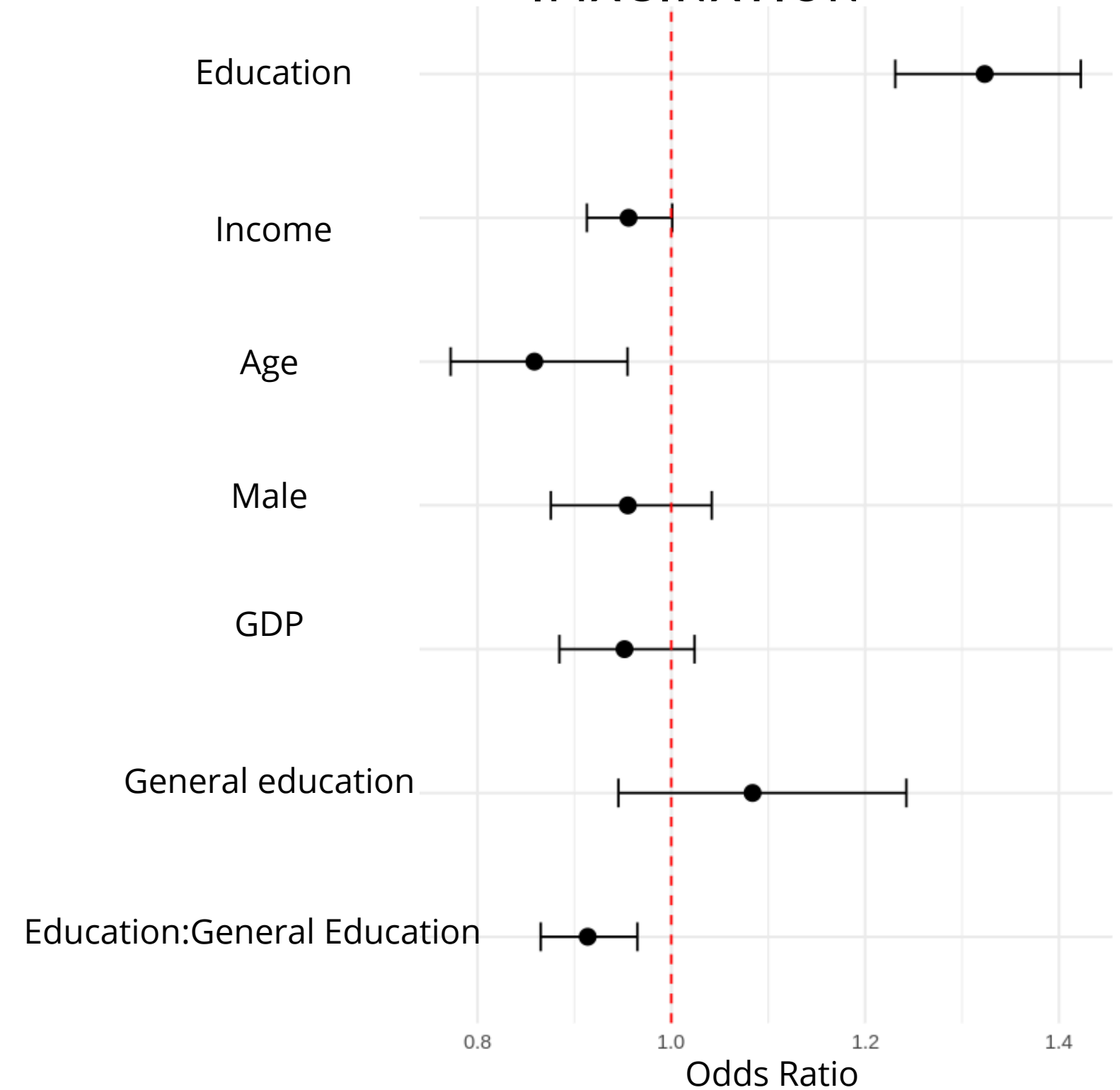
■ GENERAL EDUCATION LEVEL

H5: GENERAL EDUCATION HAS AN EFFECT ON EDUCATION EFFECT

NATIONAL PRIDE



IMAGINATION



THANK YOU !

APPENDIX

DATA

EVS-WVS 2017 dataset, N = 37771 (35 countries)

DEPENDENT VARIABLES

Trust : Generally speaking, would you say that **most people can be trusted (1)** or that you **can't be too careful (0)** in dealing with people?

Here is a list of qualities which children can be encouraged to learn at home. Which five would you say are the most desirable for a child to have?

- *Imagination* → **Openness to experience**
- *Thrift, saving money and things* → **Time discounting**
- *Not being selfish* → **Altruism**

Universalism: **Out-group mean** (how much do you trust your family, your neighbourhood, people you know personally - **in-group mean** (how much do you trust people you meet for the first time, people of another religion, people of another nationality)

National pride : How proud are you to be (Country's) citizen?
(1 - not at all proud, 2 - not very proud, 3- quite proud, 4 - very proud)

DATA

EVS-WVS 2017 dataset, N = 37771 (35 countries)

INDEPENDENT VARIABLES

Education : highest level of education attained, 9 levels from no education to doctor

Perceived income : On this card is an income scale on which 1 indicates the lowest income group and 10 the highest income group in your country. We would like to know in what group your household is.

Age : starting from 25 up to 82

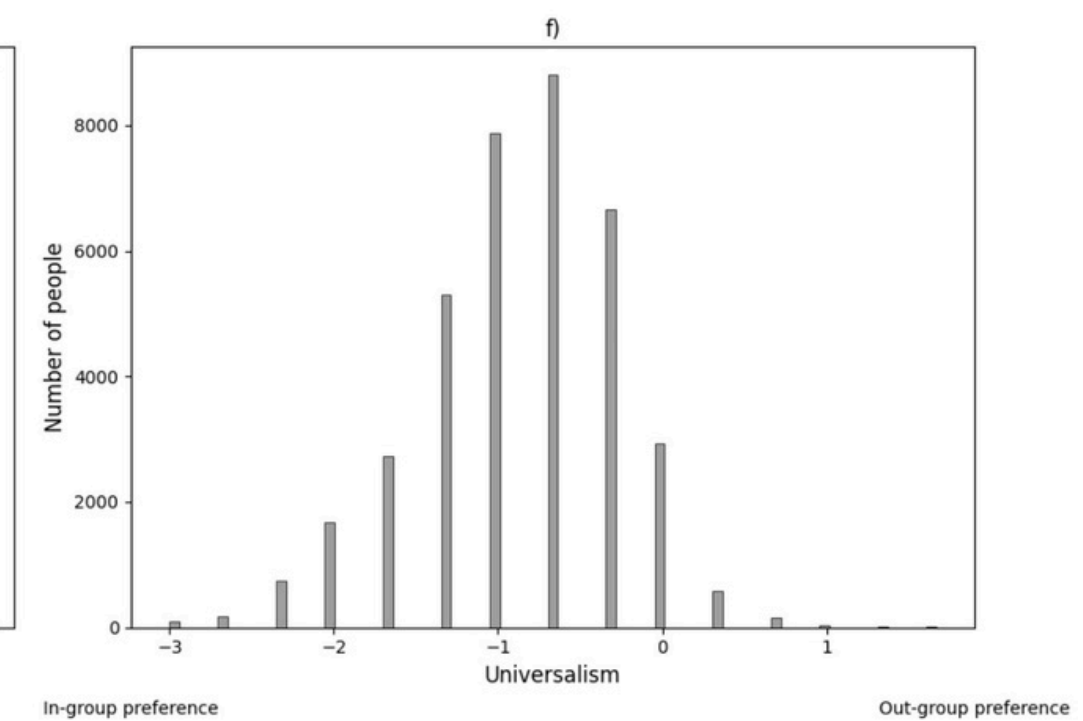
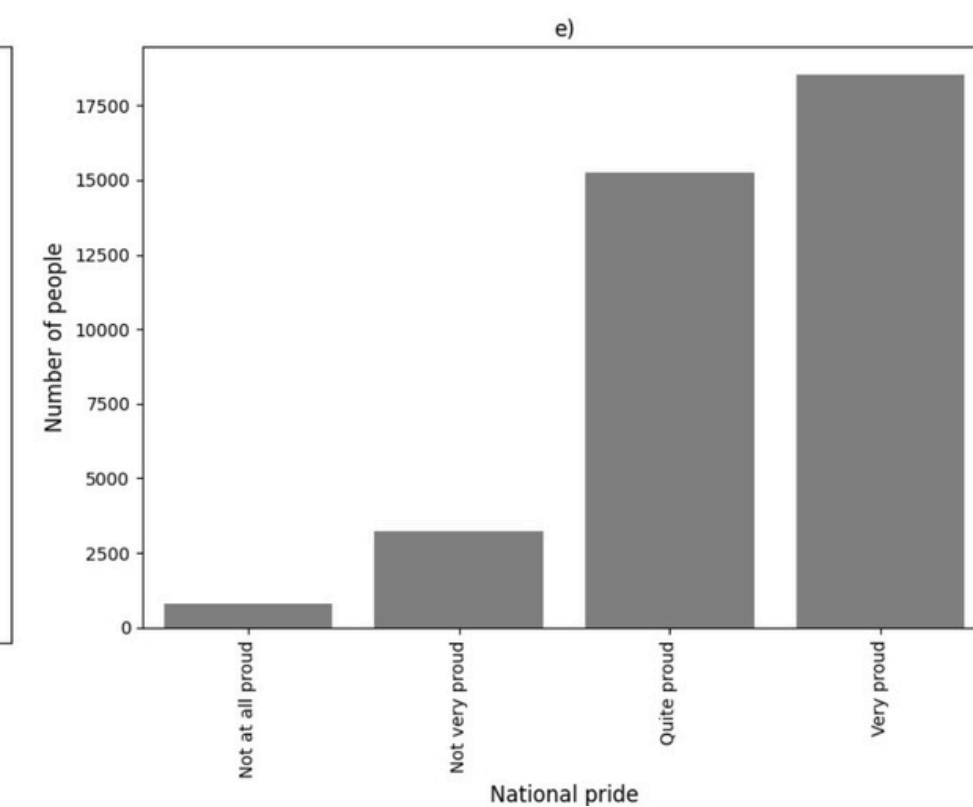
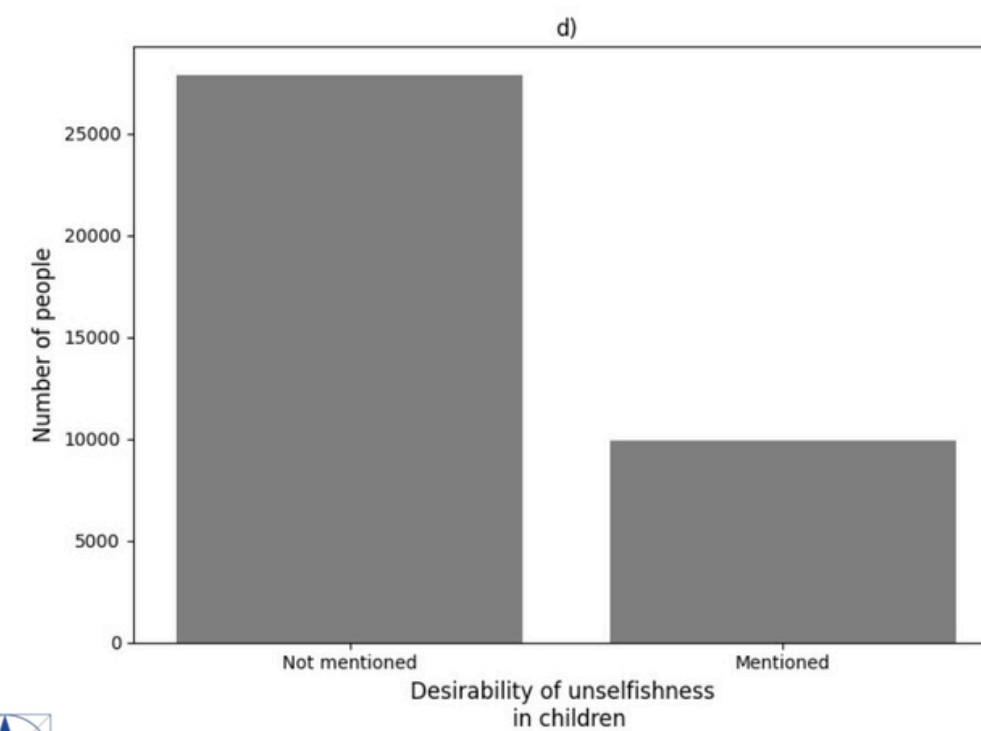
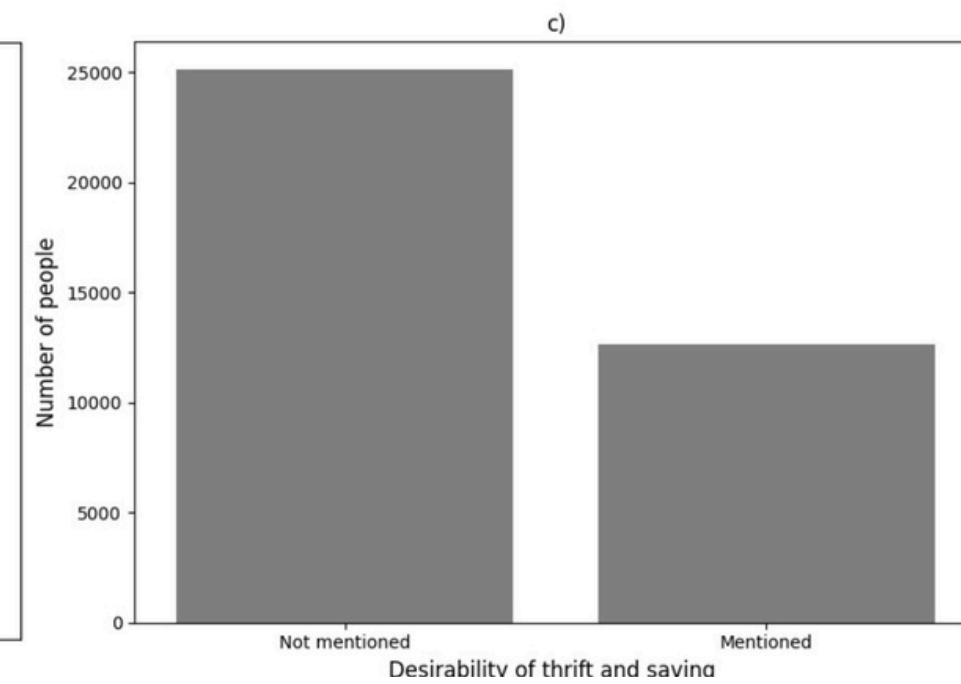
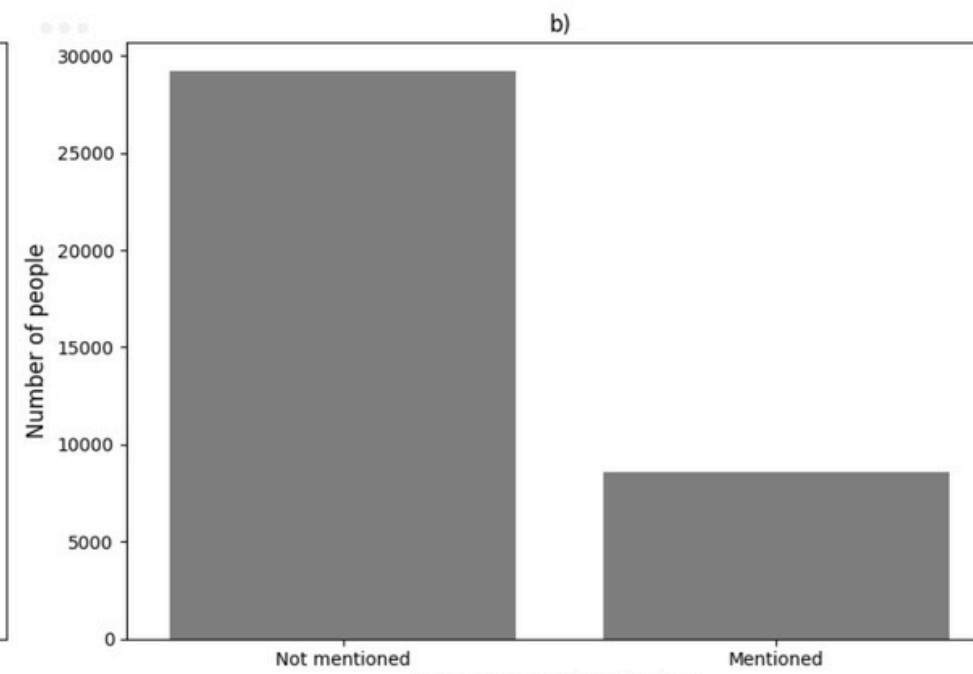
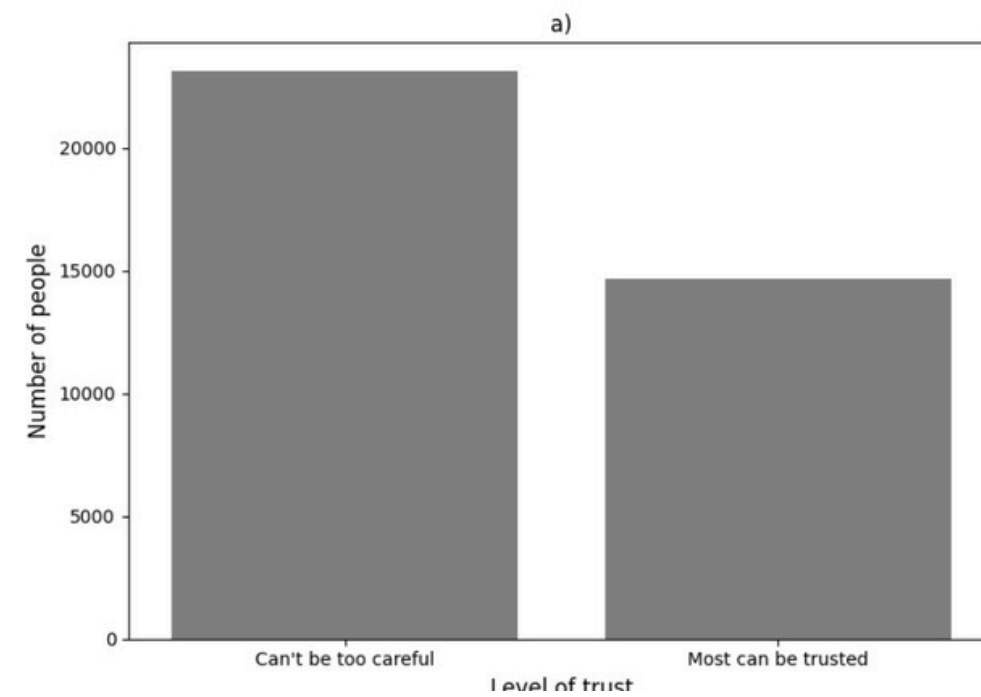
Sex : men, women



DATA

EVS-WVS 2017 dataset, N = 37771 (35 countries)

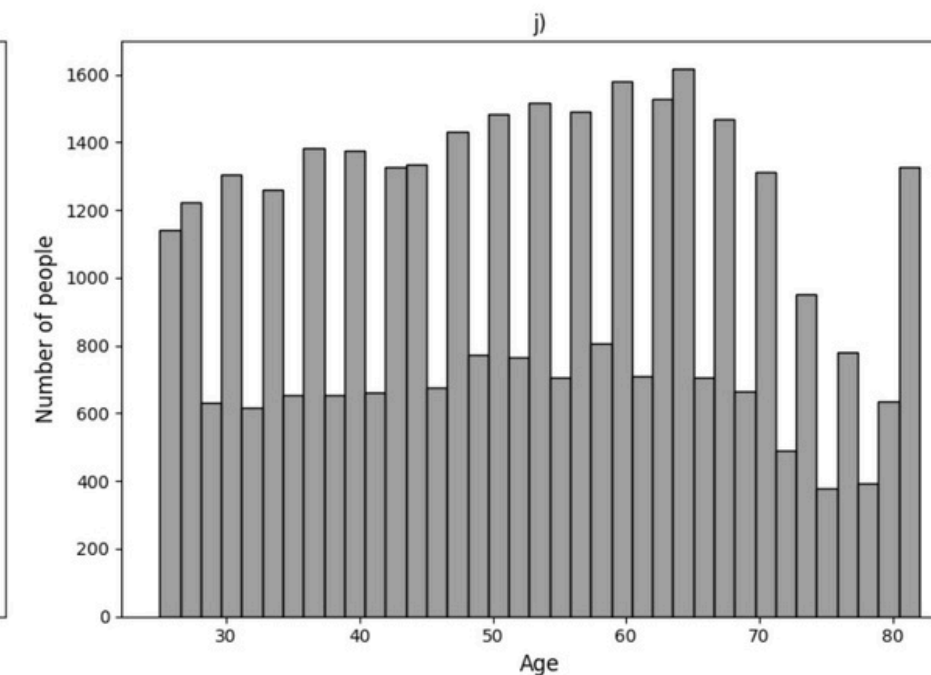
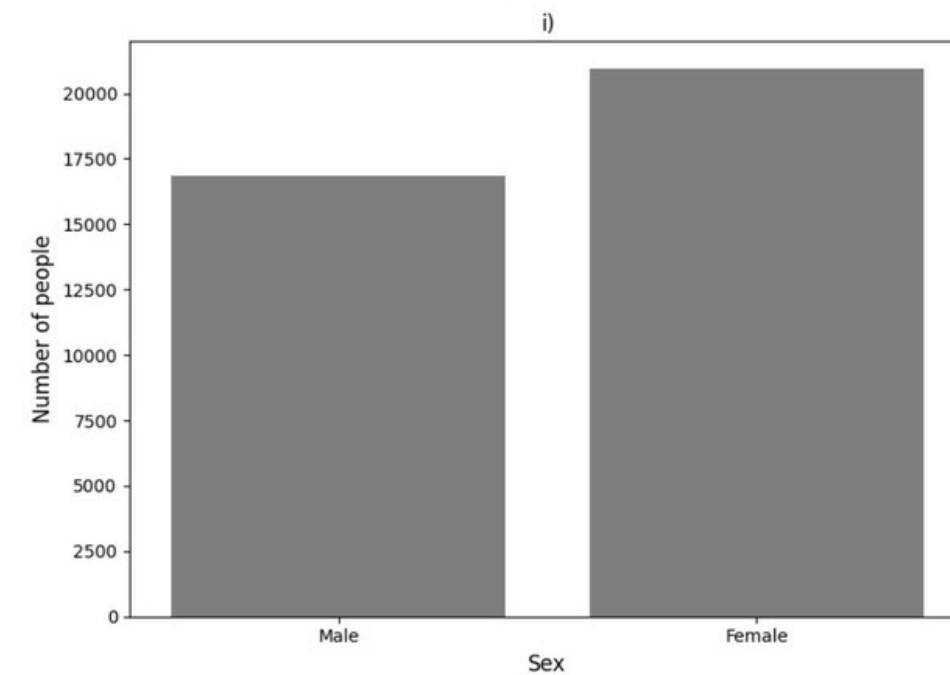
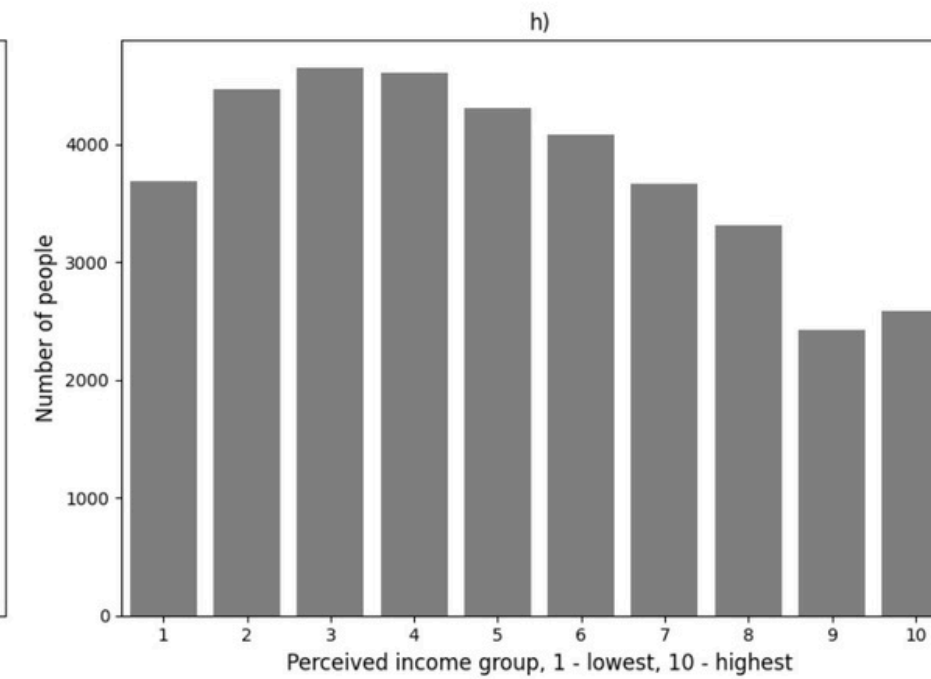
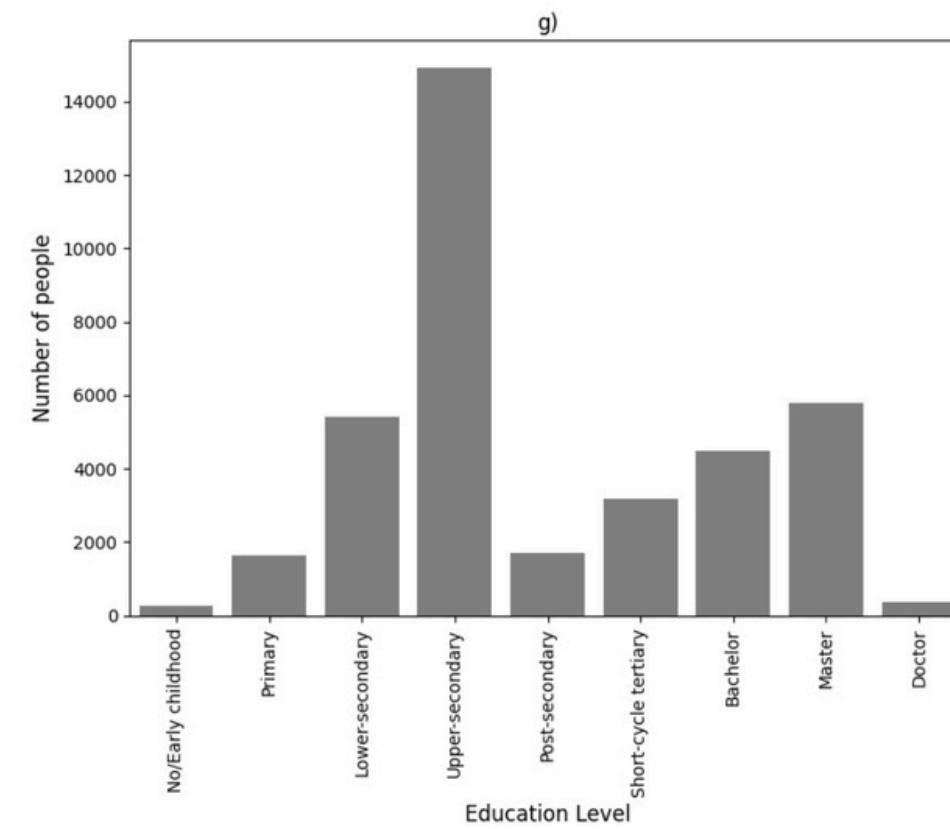
DEPENDENT VARIABLES

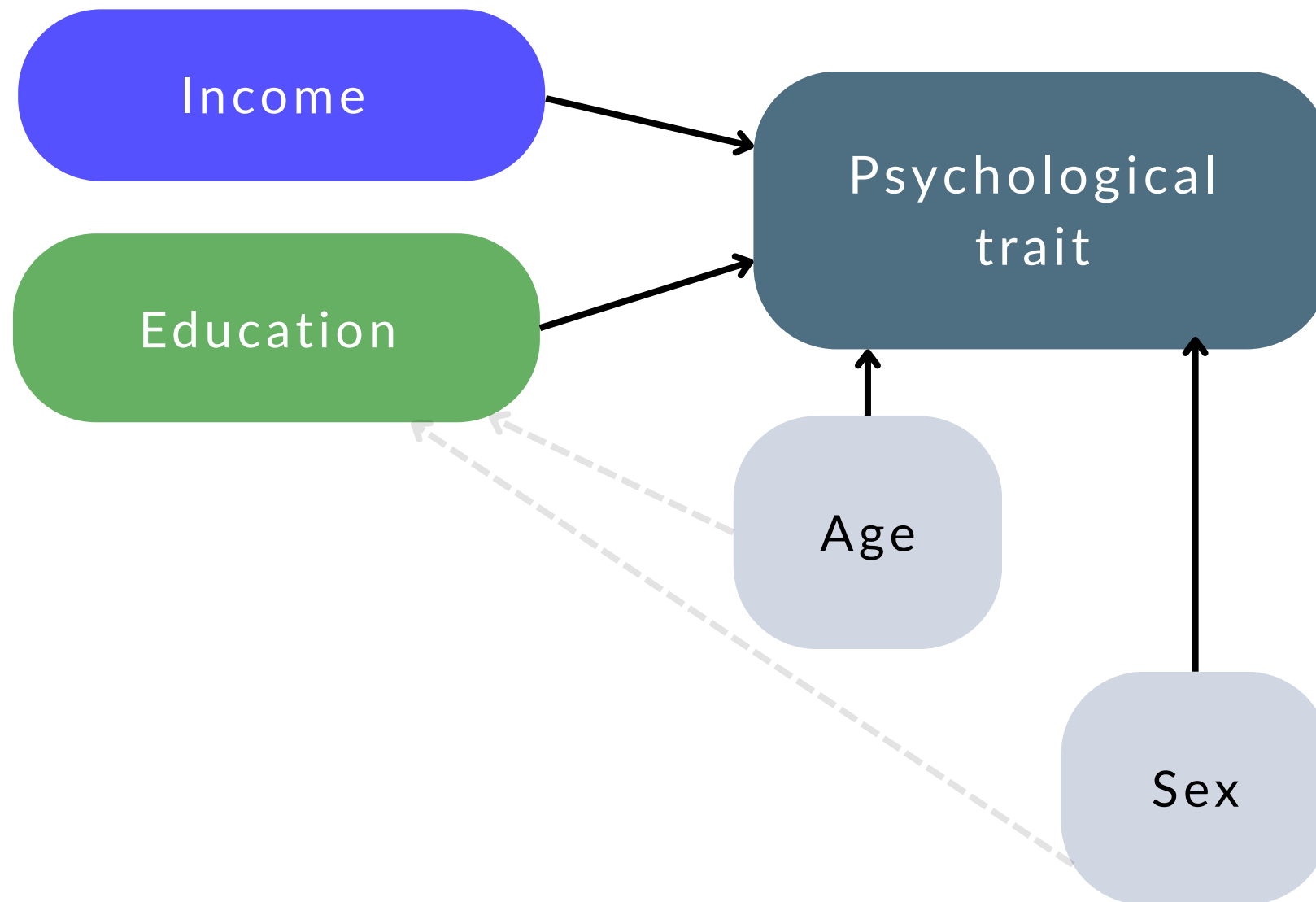


DATA

INDEPENDENT VARIABLES

$$\text{Socio-cultural attitude} = \beta_0 + \beta_1 \text{Education} + \beta_2 \text{Income} + \beta_3 \text{Age} + \beta_4 \text{Sex} + \varepsilon$$





Income

1. An impact on costs of your decisions
2. Increased stress , impaired mental abilities

Education

1. Cognitive explanation: critical spirit instead of heuristics
2. Socialisation theory: transmission of norms, values, and models of behaviour deemed to be appropriate in a given society

(Coenders and Scheepers, 2003)

H1: EDUCATION AND INCOME ARE CORRELATED WITH TRUST, ALTRUISM, OPENNESS TO EXPERIENCE, TIME DISCOUNTING, UNIVERSALISM, NATIONAL PRIDE

Logistic regression



HETEROGENEITY

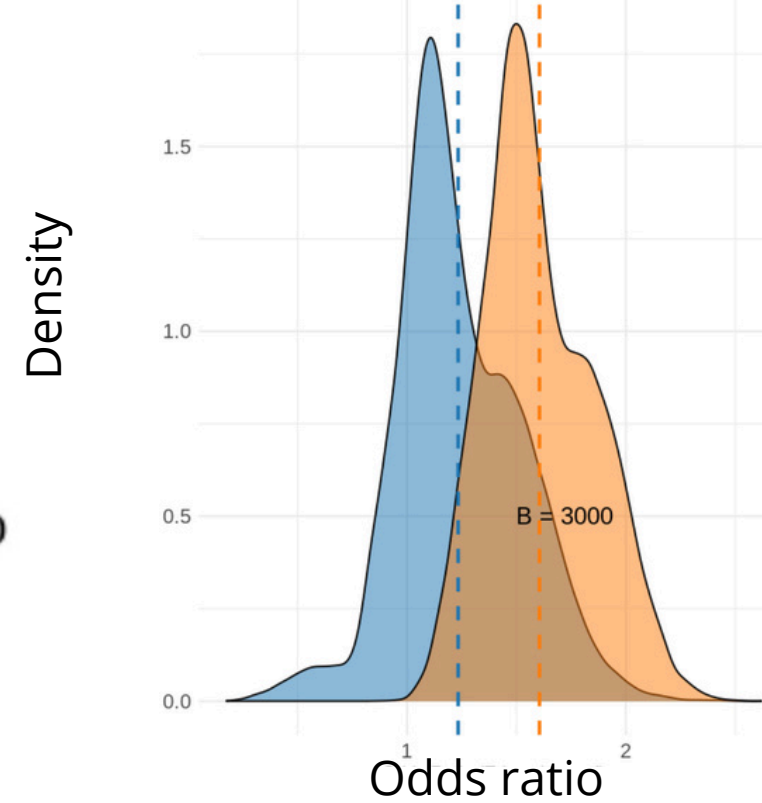
Variable	Edu Q	p-value	I ² edu	Inc Q	p-value	I ² inc
Trust	247.8	<<0.001	85.97%	121.2	<<0.001	73.63%
Unselfishness	67.3	< 0.001	50.09%	50.2	< 0.05	33.83%
Imagination	139.0	<<0.001	76.47%	50.9	< 0.05	30.43%
Thrift and saving	117.6	<<0.001	70.36%	44.0	>>0.5	12.71%
Universalism	101.3	<<0.001	67.60%	90.7	<<0.001	63.56%
National pride	192.9	<<0.001	83.74%	103.7	<0.001	67.31%

I² < 25% — low heterogeneity; 25% - 50% — moderate heterogeneity; > 50% — high heterogeneity

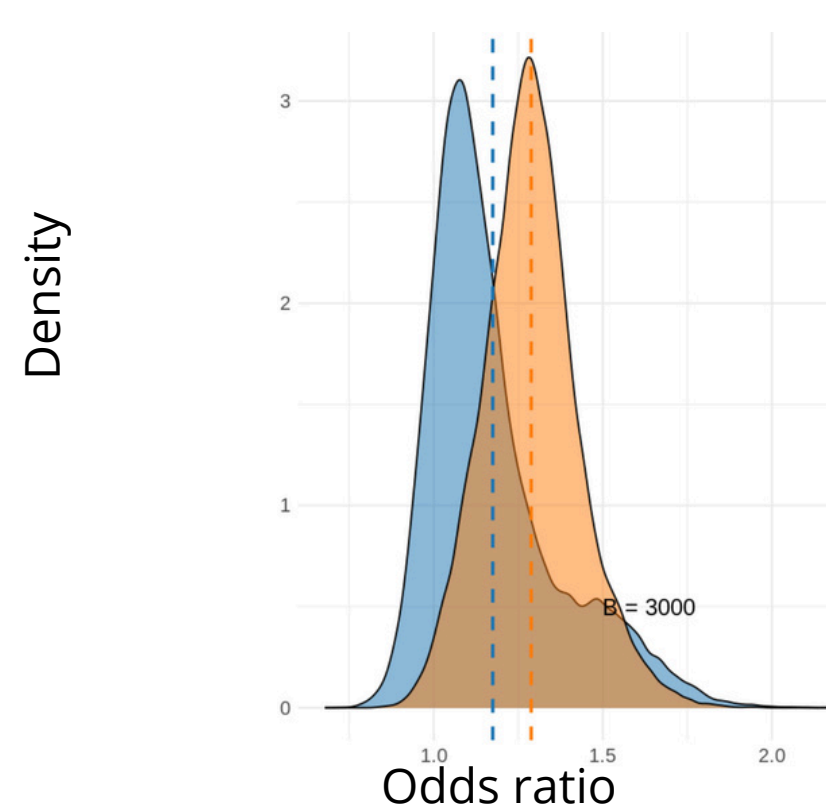
EAST VS WEST? EDUCATION

H3: THERE IS A DIFFERENCE IN EDUCATION EFFECT BETWEEN EAST AND WEST

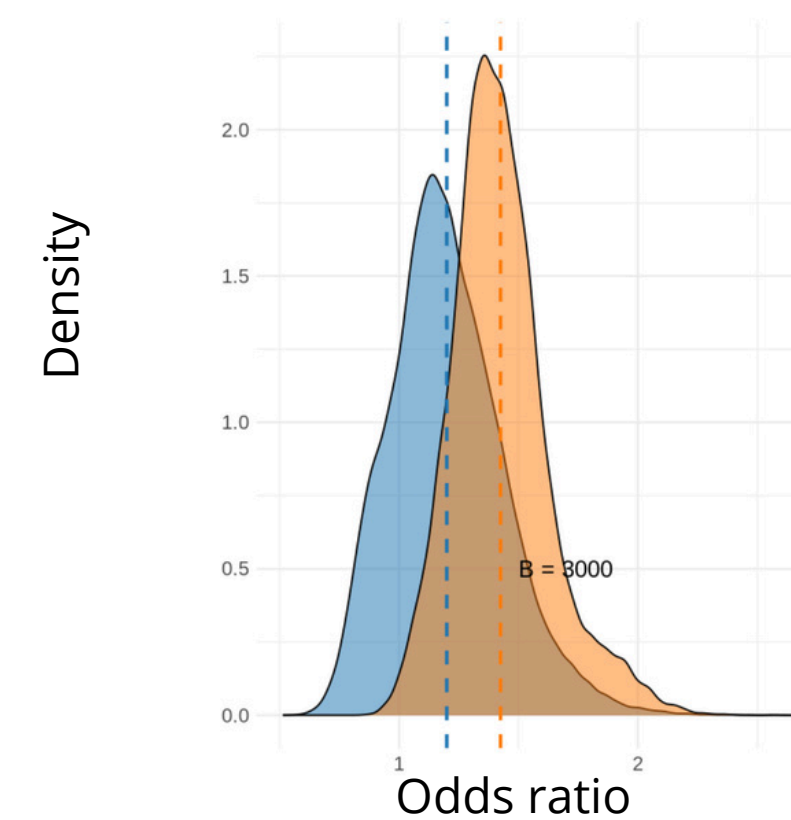
TRUST



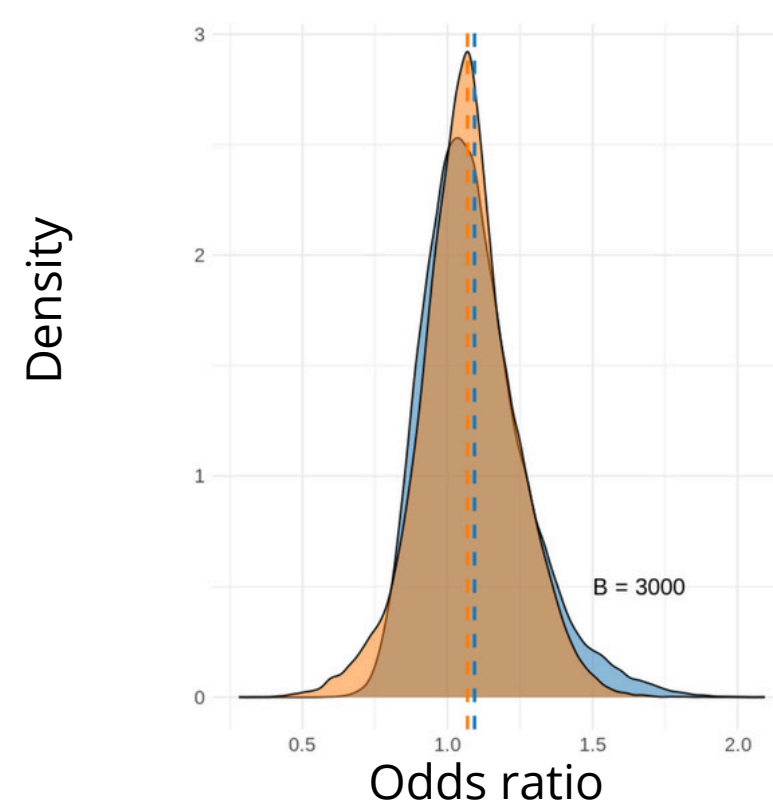
UNIVERSALISM



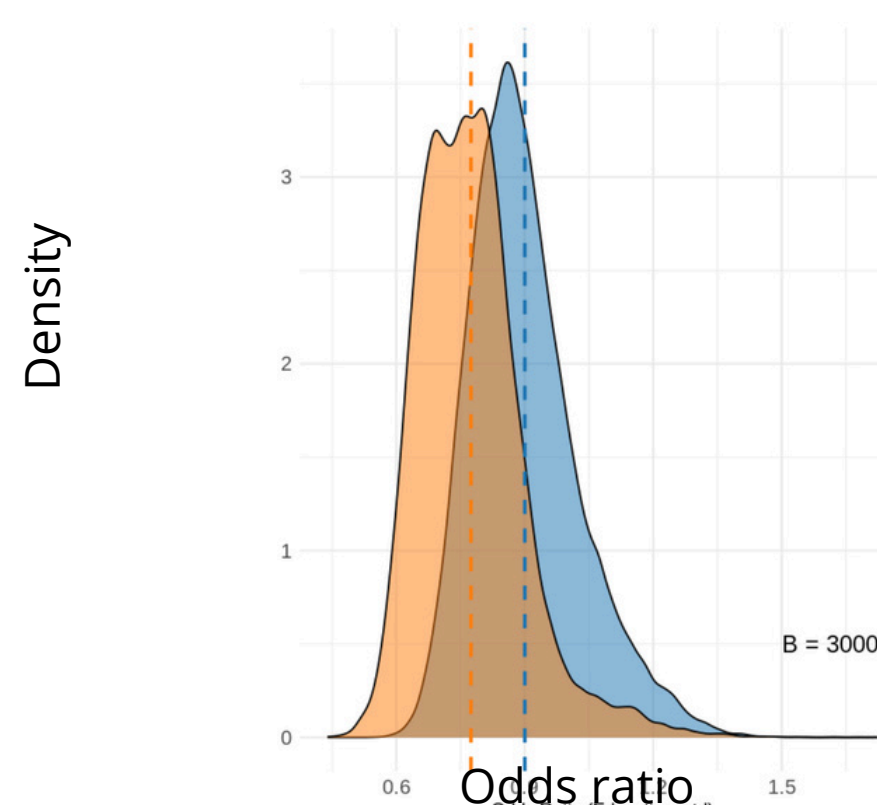
OPENNESS TO EXPERIENCE



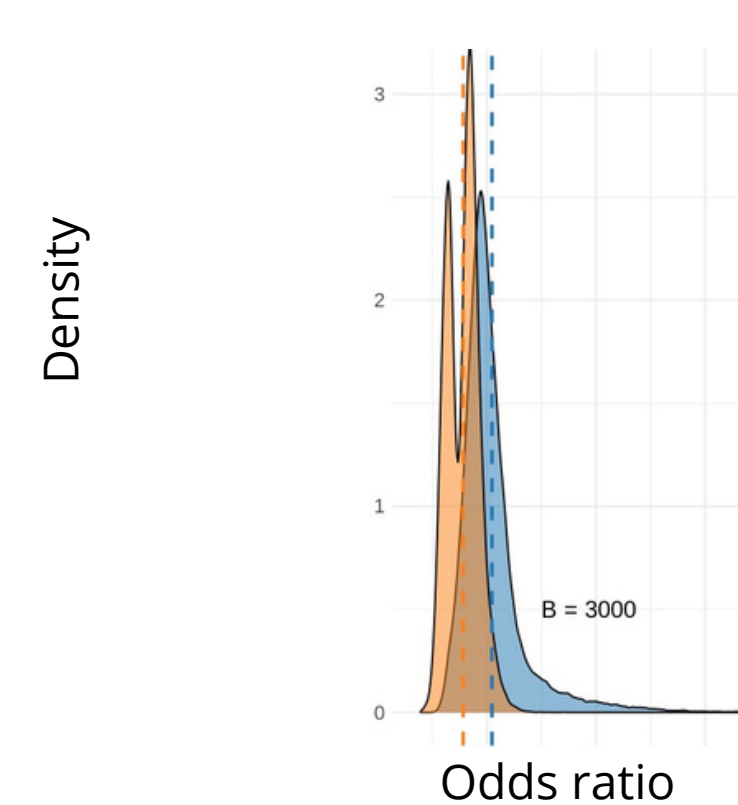
ALTRUISM



TIME DISCOUNTING



NATIONAL PRIDE



■ EXPLANATION

H4 : IN LONGER DEMOCRACIES EDUCATION EFFECT IS STRONGER

